

Mục lục

1

Math

1.1Sequences

1.2Combinatoric

1.3Number of spanning tree

1.4Series

1.5Geometry

1.6Pick's Theorem

1.7Erdos Gallai Theorem

2

Programming

2.1Pragma

3

Flow and Matching

3.1Maximum Flow (Dinic)

3.2Dinic (Template by Tran Khoi Nguyen)

3.3Min Cut (Template by Tran Khoi Nguyen)

3.4Maximum Matching (Hopcroft - Karp)

3.5Maximum Matching (Template by Tran Khoi Nguyen)

3.6Maximum Matching (O(n²))

3.7Min Cost Flow

3.8Min Cost Flow (Template by Tran Khoi Nguyen)

3.9General Graph Matching

4

Geometry

4.1Smallest Circle - Emo Welzl (Contain all points)

4.2Closest pair of points in set

4.3Manhattan MST

4.4Geometry with real number

5

Numerical algorithms

5.1SQRT For Loop

5.2Rabin Miller - Prime Checker

5.3Chinese Remain Theorem

5.4Pollard Rho - Factorialize

5.5FFT

5.6FFT (Mod 998244353)

5.7FFT Mod (Template by Tran Khoi Nguyen)

5.8Fast Walsh-Hadamard Transform (FWHT)

5.9Fast Walsh-Hadamard Transform (FWHT Base m)

5.10Count Primes

5.11Interpolation (Mod a prime)

5.12Bignum

5.13Bignum with FFT multiplication

5.14Tonelli Shanks (Find square root modulo prime)

5.15Discrete Logarithm (Find x that $a^x \equiv b \pmod m$)

5.16Primitive Root (Exist k that $g^k \equiv a \pmod n$ for all a)

5.17Discrete Root (Find x that $x^k \equiv a \pmod n$, n is a prime)

5.18Super Sieve of Primes (Code by RR)

6

Graph algorithms

6.1Twosat (2-SAT)

6.2Eulerian Path

6.3Biconnected Component Tree

7

String

7.1Palindrome Tree

7.2Suffix Array

7.3Suffix Array (Template by Tran Khoi Nguyen)

7.4Aho Corasick - Extended KMP

7.5Aho Corasick (Template by Tran Khoi Nguyen)

7.6

Suffix Tree (Template by Tran Khoi Nguyen)

19

7.7

Z Function

19

8

Data structures

19

8.1

Ordered Set

19

8.2

Fenwick Tree (With Walk on tree)

19

8.3

Convex Hull Trick (Min)

19

8.4

Dynamic Convex Hull Trick (Min)

20

8.5

SPlay Tree

20

8.6

Hashing (Template by Tran Khoi Nguyen)

21

8.7

BIT 2D (Template by Tran Khoi Nguyen)

21

8.8

Wavelet Tree

21

1

Math

1.1

Sequences

Catalan Numbers (OEIS A000108)

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}, \quad C_0 = 1$$

Recurrence:

$$C_{n+1} = \sum_{i=0}^n C_i C_{n-i}$$

Generating function:

$$C(x) = \frac{1 - \sqrt{1 - 4x}}{2x}$$

Applications (exact meanings):

- Number of valid parenthesis expressions with n pairs.
- Number of monotonic lattice paths from $(0, 0)$ to (n, n) staying below the diagonal.
- Number of triangulations of a convex polygon with $n + 2$ vertices.
- Number of full binary trees with $n + 1$ leaves.
- Number of noncrossing partitions of an n -element set.

Bernoulli Numbers (OEIS A027641)

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} B_n \frac{x^n}{n!}, \quad B_0 = 1, \quad B_1 = -\frac{1}{2}, \quad B_{2m+1} = 0 \quad (m \geq 1)$$

Key formulas:

$$\sum_{k=1}^n k^p = \frac{1}{p+1} \sum_{j=0}^p \binom{p+1}{j} B_j n^{p+1-j}$$

$$\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{2(2m)!} B_{2m}$$

Applications (precise uses):

- Compute closed-form sums of powers up to degree p .
- Evaluate Riemann zeta at even positive integers.
- Correct high-order error in Euler–Maclaurin summation for n -term truncation.

Stirling Numbers of the First Kind (signed) (OEIS A048994)

$$f_n(x) = x(x-1) \cdots (x-n+1) = \sum_{k=0}^n s(n, k) x^k$$

Recurrence:

$$s(n, k) = s(n-1, k-1) - (n-1)s(n-1, k), s(0, 0) = 1$$

Applications:

- $|s(n, k)|$: number of permutations of n elements with exactly k cycles.
- Convert falling factorial basis $f_n(x)$ to power basis x^k for degree n polynomials.

Stirling Numbers of the Second Kind (OEIS A008277)

$$x^n = \sum_{k=0}^n S(n, k) f_k(x), \quad S(n, k) = kS(n-1, k) + S(n-1, k-1), \quad S(0, 0) = 1$$

Closed form:

$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

Applications:

- Number of ways to partition an n -element labeled set into k nonempty unlabeled blocks.
- Number of surjective functions from n elements onto k labeled elements ($= k!S(n, k)$).
- Coefficients when expanding monomials x^n into falling factorial basis $f_k(x)$.

Bell Numbers (OEIS A000110)

$$B_n = \sum_{k=0}^n S(n, k), \quad \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n = e^{e^x - 1}$$

Dobinski's formula:

$$B_n = \frac{1}{e} \sum_{m=0}^{\infty} \frac{m^n}{m!}$$

Applications:

- Number of all set partitions of an n -element set.
- Number of equivalence relations on an n -element set.
- Number of ways to cluster n labeled objects into any number of groups.

Eulerian Numbers (OEIS A173018)

Let $A(n, k)$ = number of permutations of $\{1, \dots, n\}$ with exactly k ascents.

Recurrence:

$$A(n, k) = (n-k)A(n-1, k-1) + (k+1)A(n-1, k), \quad A(1, 0) = 1$$

Exponential generating function:

$$\sum_{n, k \geq 0} A(n, k) \frac{t^n}{n!} x^k = \frac{1-x}{1-xe^{t(1-x)}}$$

Applications:

- $A(n, k)$: number of permutations of length n with exactly k descents.
- Coefficients of Eulerian polynomial $A_n(x) = \sum_{k=0}^{n-1} A(n, k)x^k$.
- Expected number of descents in random permutation of length n : $(n-1)/2$.

Computation notes:

- Catalan: compute via factorials mod p .
- Stirling (1st/2nd): dynamic programming in $O(n^2)$.
- Bell: sum $S(n, k)$ over k .
- Eulerian: DP with recurrence above.
- Bernoulli: Akiyama–Tanigawa algorithm for small n .

1.2 Combinatoric

Basic: Binomial & Multinomial

$$(1-x)^{-a} = \sum_{k=0}^{\infty} \binom{a+k-1}{k} x^k \quad (|x| < 1).$$

$$\binom{-a}{k} = (-1)^k \binom{a+k-1}{k},$$

$$\sum_{k=0}^n \binom{n}{k} = 2^n, \quad \sum_{k=0}^n k \binom{n}{k} = n 2^{n-1}, \quad \sum_{k=0}^n k^2 \binom{n}{k} = n(n+1) 2^{n-2}.$$

Hockey-stick (Christmas stocking)

$$\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1} \quad (0 \leq r \leq n).$$

Vandermonde (binomial convolution)

$$\sum_k \binom{r}{k} \binom{s}{n-k} = \binom{r+s}{n} \quad (n \in \mathbb{Z}).$$

Notable corollary: $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}.$

Chu–Vandermonde (negative coefficients form)

$$\sum_{k=0}^n \binom{a}{k} \binom{b}{n-k} = \binom{a+b}{n} \quad (a, b \in \mathbb{R}),$$

use $\binom{-m}{k} = (-1)^k \binom{m+k-1}{k}$ to handle "selection with replacement".

Multinomial polynomial

$$(x_1 + \dots + x_m)^n = \sum_{k_1 + \dots + k_m = n} \binom{n}{k_1, \dots, k_m} \prod_{i=1}^m x_i^{k_i},$$

$$\binom{n}{k_1, \dots, k_m} = \frac{n!}{\prod_i k_i!}, \quad \sum_{k_1 + \dots + k_m = n} \binom{n}{k_1, \dots, k_m} = m^n.$$

Binomial transform

$$g_n = \sum_{k=0}^n \binom{n}{k} f_k \iff f_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} g_k.$$

Power form: $\sum_{k=0}^n (-1)^k \binom{n}{k} k^m = 0$ for $0 \leq m < n$.

Power sum via Bernoulli (Faulhaber)

$$\sum_{k=1}^N k^p = \frac{1}{p+1} \sum_{j=0}^p \binom{p+1}{j} B_j N^{p+1-j},$$

B_j is Bernoulli.
Special cases for small n :

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(2n+1)(n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

Permutations with fixed points

$$\left| \bigcup_{i=1}^m A_i \right| = \sum_{j \geq 1} (-1)^{j-1} \sum_{1 \leq i_1 < \dots < i_j \leq m} |A_{i_1} \cap \dots \cap A_{i_j}|.$$

Derangements (permutations with zero fixed points):

$$!0 = 1, \quad !1 = 0, \quad !m = (m-1) (! (m-1) + ! (m-2)),$$

and the inclusion–exclusion form

$$!m = m! \sum_{i=0}^m \frac{(-1)^i}{i!}.$$

Count of permutations $\pi \in S_n$ with exactly k fixed points:

$$R(n, k) = \binom{n}{k} ! (n - k),$$

Related to Stirling

Let $f_n(x) = x(x-1) \dots (x-n+1)$.

$$f_n(x) = \sum_{k=0}^n s(n, k) x^k, \quad x^n = \sum_{k=0}^n S(n, k) f_k(x).$$

Number of surjective maps $[n] \rightarrow [k]$: $k! S(n, k) = \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n.$

Some common alternating sums

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = 0 \quad (n \geq 1), \quad \sum_{k=0}^n (-1)^k \binom{r+k}{k} = (-1)^n \binom{r}{n}.$$

Central coefficient & approximation

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2, \quad \binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}.$$

1.3 Number of spanning tree

Create an $N \times N$ matrix `mat`, and for each edge $a \rightarrow b \in G$, do `mat[a][b]−`, `mat[b][b]++` (and `mat[b][a]−`, `mat[a][a]++` if G is undirected). Remove the i th row and column and take the determinant; this yields the number of directed spanning trees rooted at i (if G is undirected, remove any row/column).

1.4 Series

Basic Series (Maclaurin/Taylor around 0)

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad (\forall x), \quad \ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} \quad (|x| < 1)$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1), \quad \frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n \quad (|x| < 1)$$

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n, \quad \binom{\alpha}{n} = \frac{\alpha(\alpha-1) \dots (\alpha-n+1)}{n!} \quad (|x| < 1)$$

Trigonometric & Hyperbolic

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \quad (\forall x)$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \quad \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} \quad (\forall x)$$

Use Bernoulli to quickly remember fractional series:

$$\tan x = \sum_{m=1}^{\infty} \frac{(-1)^{m-1} 2^{2m} (2^{2m} - 1) B_{2m}}{(2m)!} x^{2m-1} \quad (|x| < \frac{\pi}{2})$$

$$x \cot x = 1 - \sum_{m=1}^{\infty} \frac{2^{2m} B_{2m}}{(2m)!} x^{2m} \quad (0 < |x| < \pi)$$

Inverse Trigonometric & Logarithmic

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} \quad (|x| \leq 1, x \neq \pm i)$$

$$\arcsin x = \sum_{n=0}^{\infty} \frac{\binom{2n}{n}}{4^n (2n+1)} x^{2n+1} \quad (|x| < 1), \quad \arccos x = \frac{\pi}{2} - \arcsin x$$

$$\ln(1-x) = - \sum_{n=1}^{\infty} \frac{x^n}{n} \quad (|x| < 1), \quad \ln \frac{1+x}{1-x} = 2 \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1} \quad (|x| < 1)$$

Generating Series & Fast Combinatorics

$$\frac{1}{(1-x)^\alpha} = \sum_{n=0}^{\infty} \binom{\alpha+n-1}{n} x^n \quad (|x| < 1)$$

$$\ln \frac{1}{1-x} = \sum_{n=1}^{\infty} \frac{x^n}{n} \quad (|x| < 1), \quad \frac{x}{1-e^{-x}} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n \quad (|x| < 2\pi)$$

Infinite Products (useful for checking/transformation)

$$\sin(\pi x) = \pi x \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2} \right), \quad \frac{\sinh(\pi x)}{\pi x} = \prod_{n=1}^{\infty} \left(1 + \frac{x^2}{n^2} \right)$$

1.5 Geometry

Distance from a point to a line

Given a point $P(x_0, y_0)$ and a line $d : ax + by + c = 0$:

$$d(P, d) = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}.$$

Distance between two lines

Two parallel lines: $d_1 : ax + by + c_1 = 0, d_2 : ax + by + c_2 = 0$:

$$d(d_1, d_2) = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}.$$

1.6 Pick's Theorem

Pick's Theorem relates the area of a polygon with integer coordinates: Let S = area of polygon, I = number of integer points inside polygon, B = number of integer points on polygon boundary
Then: $S = I + \frac{B}{2} - 1$

1.7 Erdos Gallai Theorem

A simple graph with node degrees $d_1 \geq \dots \geq d_n$ exists iff $d_1 + \dots + d_n$ is even and for every $k = 1 \dots n$:

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k).$$

2 Programming

2.1 Pragma

```
#pragma GCC optimize("O3,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,popcnt,lzcnt")
```

3 Flow and Matching

3.1 Maximum Flow (Dinic)

// In case we need to find Maximum flow of network with both minimum capacity and maximum capacity, let s and t* be virtual source and virtual sink.*

/
Then, each edge (u->v) with lower cap l and upper cap r will be changed in to 3 edge:*

*- u->v with capacity r-l
- u->t* with capacity l
- s*->v with capacity l
/

*// We need add one other edge t->s with capacity Inf
// Maximum Flow on original graph is the Maximum Flow on new graph
: s*->t**

```
struct Edge
{
    int u, v;
    ll c;
    Edge() {}
    Edge(int u, int v, ll c)
    {
        this->u = u;
        this->v = v;
        this->c = c;
    }
};
struct Dinic
{
    const ll Inf = 1e17;
    vector<vector<int>>> adj;
    vector<vector<int>>::iterator> cur;
    vector<Edge> s;
    vector<int> h;
    int sink, t;
    int n;
    Dinic(int n)
    {
        this->n = n;
        adj.resize(n + 1);
        h.resize(n + 1);
        cur.resize(n + 1);
        s.reserve(n);
    }
    void AddEdge(int u, int v, ll c)
    {
        s.emplace_back(u, v, c);
        adj[u].push_back(s.size() - 1);
    }
};
```

```
s.emplace_back(v, u, 0);
adj[v].push_back(s.size() - 1);
}
bool BFS()
{
    fill(h.begin(), h.end(), n + 2);
    queue<int> pq;
    h[t] = 0;
    pq.emplace(t);
    while (pq.size())
    {
        int c = pq.front();
        pq.pop();
        for (auto i : adj[c])
            if (h[s[i ^ 1].u] == n + 2 && s[i ^ 1].c != 0)
            {
                h[s[i ^ 1].u] = h[c] + 1;
                if (s[i ^ 1].u == sink)
                    return true;
                pq.emplace(s[i ^ 1].u);
            }
    }
    return false;
}
ll DFS(int v, ll flowin)
{
    if (v == t)
        return flowin;
    ll flowout = 0;
    for (; cur[v] != adj[v].end(); ++cur[v])
    {
        int i = *cur[v];
        if (h[s[i].v] + 1 != h[v] || s[i].c == 0)
            continue;
        ll q = DFS(s[i].v, min(flowin, s[i].c));
        flowout += q;
        if (flowin != Inf)
            flowin -= q;
        s[i].c -= q;
        s[i ^ 1].c += q;
        if (flowin == 0)
            break;
    }
    return flowout;
}
void BlockFlow(ll &flow)
{
    for (int i = 1; i <= n; ++i)
        cur[i] = adj[i].begin();
    flow += DFS(sink, Inf);
}
ll MaxFlow(int s, int t)
{
    this->sink = s;
    this->t = t;
    ll flow = 0;
    while (BFS())
        BlockFlow(flow);
    return flow;
};
```

3.2 Dinic (Template by Tran Khoi Nguyen)

```
struct Edge
{
    int u, v;
    ll cap, flow;
    Edge() {}
    Edge(int u, int v, ll cap) : u(u), v(v), cap(cap), flow(0) {}
};
struct Dinic
{
    int N;
    vector<Edge> E;
    vector<vector<int>>> g;
```

```
vector<int> d, pt;
map<ll, map<ll, ll>> mp;
Dinic(int N) : N(N), E(0), g(N), d(N), pt(N) {}

void AddEdge(int u, int v, ll cap)
{
    if (u != v)
    {
        E.emplace_back(u, v, cap);
        g[u].emplace_back(E.size() - 1);
        E.emplace_back(v, u, 0);
        g[v].emplace_back(E.size() - 1);
    }
}

bool BFS(int S, int T)
{
    queue<int> q({S});
    fill(d.begin(), d.end(), N + 1);
    d[S] = 0;
    while (!q.empty())
    {
        int u = q.front();
        q.pop();
        if (u == T)
            break;
        for (int k : g[u])
        {
            Edge &e = E[k];
            if (e.flow < e.cap && d[e.v] > d[e.u] + 1)
            {
                d[e.v] = d[e.u] + 1;
                q.emplace(e.v);
            }
        }
    }
    return d[T] != N + 1;
}

ll DFS(int u, int T, ll flow = -1)
{
    if (u == T || flow == 0)
        return flow;
    for (int &i = pt[u]; i < g[u].size(); ++i)
    {
        Edge &e = E[g[u][i]];
        Edge &oe = E[g[u][i] ^ 1];
        if (d[e.v] == d[e.u] + 1)
        {
            ll amt = e.cap - e.flow;
            if (flow != -1 && amt > flow)
                amt = flow;
            if (ll pushed = DFS(e.v, T, amt))
            {
                e.flow += pushed;
                oe.flow -= pushed;
                return pushed;
            }
        }
    }
    return 0;
}

ll MaxFlow(int S, int T)
{
    ll total = 0;
    while (BFS(S, T))
    {
        fill(pt.begin(), pt.end(), 0);
        while (ll flow = DFS(S, T))
            total += flow;
    }
    return total;
};
```

3.3 Min Cut (Template by Tran Khoi Nguyen)

```

typedef long long LL;

struct Edge
{
    int u, v;
    LL cap, flow;
    Edge() {}
    Edge(int u, int v, LL cap) : u(u), v(v), cap(cap), flow(0) {}
};

bool chk[maxn];
vector<pll> mincut;
struct Dinic
{
    int N;
    vector<Edge> E;
    vector<vector<int>> g;
    vector<int> d, pt;

    Dinic(int N) : N(N), E(0), g(N), d(N), pt(N) {}

    void AddEdge(int u, int v, LL cap)
    {
        // cout <<u<<" "<<v<<" "<<cap<<endl;
        if (u != v)
        {
            E.emplace_back(u, v, cap);
            g[u].emplace_back(E.size() - 1);
            E.emplace_back(v, u, 0);
            g[v].emplace_back(E.size() - 1);
        }
    }

    bool BFS(int S, int T)
    {
        queue<int> q({S});
        fill(d.begin(), d.end(), N + 1);
        d[S] = 0;
        while (!q.empty())
        {
            int u = q.front();
            q.pop();
            if (u == T)
                break;
            for (int k : g[u])
            {
                Edge &e = E[k];
                if (e.flow < e.cap && d[e.v] > d[e.u] + 1)
                {
                    d[e.v] = d[e.u] + 1;
                    q.emplace(e.v);
                }
            }
        }
        return d[T] != N + 1;
    }

    LL DFS(int u, int T, LL flow = -1)
    {
        if (u == T || flow == 0)
            return flow;
        for (int &i = pt[u]; i < g[u].size(); ++i)
        {
            Edge &e = E[g[u][i]];
            Edge &oe = E[g[u][i] ^ 1];
            if (d[e.v] == d[e.u] + 1)
            {
                LL amt = e.cap - e.flow;
                if (flow != -1 && amt > flow)
                    amt = flow;
                if (LL pushed = DFS(e.v, T, amt))
                {
                    e.flow += pushed;
                    oe.flow -= pushed;
                    return pushed;
                }
            }
        }
    }
};

```

```

        return 0;
    }
    void dfs1(ll u)
    {
        // cout <<u<<endl;
        chk[u] = 1;
        for (int k : g[u])
        {
            Edge e = E[k];
            if (e.cap - e.flow > 0)
            {
                ll nxt = e.v;
                if (!chk[nxt])
                    dfs1(nxt);
            }
        }
    }
    void find_mincut(ll S)
    {
        dfs1(S);
        for (int i = 0; i < E.size(); i += 2)
        {
            auto p = E[i];
            if (chk[p.u] && !chk[p.v])
            {
                mincut.pb(make_pair(p.u, p.v));
            }
        }
    }

    LL MaxFlow(int S, int T)
    {
        LL total = 0;
        while (BFS(S, T))
        {
            fill(pt.begin(), pt.end(), 0);
            while (LL flow = DFS(S, T))
                total += flow;
        }
        return total;
    }
};

```

3.4 Maximum Matching (Hopcroft - Karp)

```

// Trace to find vertex cover and independence set
/*
    Y* = Set of vertices y such that exist an argument path
        from y to a vertex x which isn't matched
    X* = Set of matched vertices x that x isn't matched with a
        vertex in Y*

    (X* v Y*) is vertex cover
*/

struct Hopcroft_Karp
{
    const int NoMatch = -1;
    vector<int> h, S, match;
    vector<vector<int>> adj;
    int nx, ny;
    bool found;
    Hopcroft_Karp(int nx = 0, int ny = 0)
    {
        this->nx = nx;
        this->ny = ny;
        S.reserve(nx);
        h.resize(ny + 5);
        adj.resize(nx + 5);
        match.resize(ny + 5, NoMatch);
    }

    void Clear()
    {
        for (int i = 1; i <= nx; ++i)
            adj[i].clear();
        S.clear();
        fill(match.begin(), match.end(), NoMatch);
    }
};

```

```

    }
    void AddEdge(int x, int y)
    {
        adj[x].emplace_back(y);
    }
    bool BFS()
    {
        fill(h.begin(), h.end(), 0);
        queue<int> q;
        for (auto x : S)
            for (auto i : adj[x])
                if (h[i] == 0)
                {
                    q.emplace(i);
                    h[i] = 1;
                }
        while (q.size())
        {
            int x, ypop = q.front();
            q.pop();
            if ((x = match[ypop]) == NoMatch)
                return true;
            for (auto i : adj[x])
                if (h[i] == 0)
                {
                    h[i] = h[ypop] + 1;
                    q.emplace(i);
                }
        }
        return false;
    }
    void dfs(int v, int lv)
    {
        for (auto i : adj[v])
            if (h[i] == lv + 1)
            {
                if (match[i] == NoMatch)
                    found = 1;
                else
                    dfs(match[i], lv + 1);
                if (found)
                {
                    match[i] = v;
                    return;
                }
            }
    }
    int MaxMatch()
    {
        int ans(0);
        for (int i = 1; i <= nx; ++i)
            S.emplace_back(i);
        while (BFS())
        {
            for (int i = S.size() - 1; ~i; --i)
            {
                found = 0;
                dfs(S[i], 0);
                if (found)
                {
                    ++ans;
                    S[i] = S.back();
                    S.pop_back();
                }
            }
        }
        return ans;
    }
};

```

3.5 Maximum Matching (Template by Tran Khoi Nguyen)

```

struct bipartite_graph
{
    int nx, ny;
    vector<vector<int>> gr;
};

```

```

vector<int> match, x_not_matched;

vector<int> level;

bool found;

void init(int _nx, int _ny)
{
    nx = _nx;
    ny = _ny;
    gr.assign(nx, vector<int>());
    x_not_matched.resize(nx);
    for (int i = 0; i < nx; ++i)
        x_not_matched[i] = i;
    match.assign(ny, -1);
    level.resize(ny);
}

void add_edge(int x, int y)
{
    gr[x].push_back(y);
}

bool bfs()
{
    fill(level.begin(), level.end(), 0);
    queue<int> q;
    for (auto &x : x_not_matched)
        for (auto &y : gr[x])
            if (level[y] == 0)
            {
                level[y] = 1;
                q.push(y);
            }
    while (!q.empty())
    {
        int ypop = q.front();
        q.pop();
        int x = match[ypop];
        if (x == -1)
            return true;
        for (auto &y : gr[x])
            if (level[y] == 0)
            {
                level[y] = level[ypop] + 1;
                q.push(y);
            }
    }
    return false;
}

void dfs(int x, int lv)
{
    for (auto &y : gr[x])
        if (level[y] == lv + 1)
        {
            level[y] = 0;
            if (match[y] == -1)
                found = true;
            else
                dfs(match[y], lv + 1);
            if (found)
            {
                match[y] = x;
                return;
            }
        }
}

int max_matching()
{
    int res = 0;
    while (bfs())
    {
        for (int i = sz(x_not_matched) - 1; i >= 0; --i)
        {
            found = false;
            dfs(x_not_matched[i], 0);

```

```

        if (found)
        {
            ++res;
            x_not_matched[i] = x_not_matched.back();
            x_not_matched.pop_back();
        }
    }
    return res;
}
} man;

```

3.6 Maximum Matching ($O(n^2)$)

```

// start from 1
// O(n^2)
struct Maximum_matching
{
    int nx, ny, t;
    vector<int> Visited, match;
    vector<vector<int>> a;

    Maximum_matching(int nx = 0, int ny = 0)
    {
        Assign(nx, ny);
    }

    void Assign(int nx, int ny)
    {
        this->nx = nx;
        this->ny = ny;
        t = 0;
        Visited.assign(nx + 5, 0);
        match.assign(ny + 5, 0);
        a.resize(nx + 5, {});
    }

    void AddEdge(int x, int y)
    {
        a[x].emplace_back(y);
    }

    bool visit(int u)
    {
        if (Visited[u] != t)
            Visited[u] = t;
        else
            return false;

        for (int i = 0; i < a[u].size(); i++)
        {
            int v = a[u][i];
            if (!match[v] || visit(match[v]))
            {
                match[v] = u;
                return true;
            }
        }
        return false;
    }

    int MaxMatch()
    {
        int ans(0);

        for (int i = 1; i <= nx; i++)
        {
            t++;
            ans += visit(i);
        }

        return ans;
    }
};

```

3.7 Min Cost Flow

```

struct Edge
{
    int u, v;
    ll c, w;
    Edge(const int &u, const int &v, const ll &c, const ll &w) : u(u), v(v), c(c), w(w) {}
};

struct MaxFlowMinCost
{
    const ll Inf = 1e17;
    int n, source, sink;
    vector<ll> d;
    vector<int> par;
    vector<bool> inqueue;
    vector<Edge> s;
    vector<vector<int>> adj;
    MaxFlowMinCost(int n)
    {
        this->n = n;
        s.reserve(n * 2);
        d.resize(n + 5);
        inqueue.resize(n + 5);
        par.resize(n + 5);
        adj.resize(n + 5);
    }

    void AddEdge(int u, int v, ll c, ll w)
    {
        s.emplace_back(u, v, c, w);
        adj[u].emplace_back(s.size() - 1);
        s.emplace_back(v, u, 0, -w);
        adj[v].emplace_back(s.size() - 1);
    }

    bool SPFA()
    {
        fill(d.begin(), d.end(), Inf);
        fill(par.begin(), par.end(), s.size());
        fill(inqueue.begin(), inqueue.end(), 0);
        d[sink] = 0;
        queue<int> q;
        q.emplace(sink);
        inqueue[sink] = 1;
        while (q.size())
        {
            int c = q.front();
            inqueue[c] = 0;
            q.pop();
            for (auto i : adj[c])
                if (s[i ^ 1].c > 0 && d[s[i].v] > d[c] + s[i ^ 1].w)
                {
                    par[s[i].v] = i ^ 1;
                    d[s[i].v] = d[c] + s[i ^ 1].w;
                    if (!inqueue[s[i].v])
                    {
                        q.emplace(s[i].v);
                        inqueue[s[i].v] = 1;
                    }
                }
        }
        return (d[source] < Inf);
    }

    pair<ll, ll> MaxFlow(int so, int t, ll k)
    {
        source = so;
        sink = t;
        ll Flow(0), cost(0);
        while (k && SPFA())
        {
            ll q(Inf);
            int v = source;
            while (v != sink)
            {
                q = min(q, s[par[v]].c);
                v = s[par[v]].v;
            }

            q = min(q, k);

```

```

    cost += d[source] * q;
    Flow += q;
    k -= q;

    v = source;
    while (v != sink)
    {
        s[par[v]].c -= q;
        s[par[v] ^ 1].c += q;
        v = s[par[v]].v;
    }
}

return {Flow, cost};
};

```

3.8 Min Cost Flow (Template by Tran Khoi Nguyen)

```
namespace MIN_COST_MAX_FLOW
```

```

{
#define REP(i, n) for (int i = 0, i##_len = (n); i < i##_len; ++i)
#define EACH(i, c) for (__typeof((c).begin()) i = (c).begin(), i##_end = (c).end(); i != i##_end; ++i)
template <class T>
inline void amin(T &x, const T &y)
{
    if (y < x)
        x = y;
}
template <class T>
inline void amax(T &x, const T &y)
{
    if (x < y)
        x = y;
}
typedef vector<int> VI;
typedef ll Flow;
typedef ll Cost;
const Flow FLOW_INF = 1LL << 60;
const Cost COST_INF = 1LL << 60;

const int SIZE = 65;
vector<pair<Cost, int>> B[SIZE];
Cost last;
int sz;

int bsr(Cost c)
{
    if (c == 0)
        return 0;
    return __lg(c) + 1;
}

void init()
{
    last = sz = 0;
    REP(i, SIZE)
        B[i].clear();
}

void push(Cost cst, int v)
{
    assert(cst >= last);
    sz++;
    B[bsr(cst ^ last)].emplace_back(cst, v);
}

pair<Cost, int> pop_min()
{
    assert(sz);
    if (B[0].empty())
    {
        int k = 1;
        while (k < SIZE && B[k].empty())
            k++;
        assert(k < SIZE);
    }
}

```

```

    last = B[k][0].first;
    EACH(e, B[k])
        amin(last, e->first);
    EACH(e, B[k])
        B[bsr(e->first ^ last)].push_back(*e);
    B[k].clear();
}

assert(B[0].size());
pair<Cost, int> ret = B[0].back();
B[0].pop_back();
sz--;
return ret;
}

struct MinCostMaxFlow
{
    struct Edge
    {
        int dst;
        Cost cst;
        Flow cap;
        int rev;
    };
    typedef vector<vector<Edge>> Graph;
    Graph G;
    bool negative_edge;
    MinCostMaxFlow(int N) : G(N)
    {
        negative_edge = false;
    }

    void add_edge(int u, int v, Cost x, Flow f)
    {
        if (u == v)
            return;
        if (x < 0)
            negative_edge = true;
        G[u].push_back((Edge){
            v, x, f, (int)G[v].size()});
        G[v].push_back((Edge){
            u, -x, 0, (int)G[u].size() - 1});
    }

    void bellman_ford(int s, vector<Cost> &h)
    {
        fill(h.begin(), h.end(), COST_INF);
        vector<bool> in(G.size());
        h[s] = 0;
        in[s] = true;
        VI front, back;
        front.push_back(s);
        while (1)
        {
            if (front.empty())
            {
                if (back.empty())
                    return;
                swap(front, back);
                reverse(front.begin(), front.end());
            }
            int v = front.back();
            front.pop_back();
            in[v] = false;
            EACH(e, G[v])
            {
                if (e->cap)
                {
                    int w = e->dst;
                    if (h[w] > h[v] + e->cst)
                    {
                        h[w] = h[v] + e->cst;
                        if (!in[w])
                        {
                            back.push_back(w);
                            in[w] = true;
                        }
                    }
                }
            }
        }
    }
}

```

```

}

Flow flow;
Cost cost;
Flow solve(int s, int t, Flow limit = FLOW_INF)
{
    flow = 0;
    cost = 0;
    vector<Cost> len(G.size()), h(G.size());
    if (negative_edge)
        bellman_ford(s, h);

    vector<int> prev(G.size()), prev_num(G.size());
    while (limit > 0)
    {
        init();
        push(0, s);
        fill(len.begin(), len.end(), COST_INF);
        fill(prev.begin(), prev.end(), -1);
        len[s] = 0;
        while (sz)
        {
            pair<Cost, int> p = pop_min();
            Cost cst = p.first;
            int v = p.second;
            if (cst > len[v])
                continue;
            for (int i = 0; i < (int)G[v].size(); i++)
            {
                const Edge &f = G[v][i];
                Cost tmp = len[v] + f.cst + h[v] - h[f.dst];
                if (f.cap > 0 && len[f.dst] > tmp)
                {
                    len[f.dst] = tmp;
                    push(tmp, f.dst);
                    prev[f.dst] = v;
                    prev_num[f.dst] = i;
                }
            }
        }

        if (prev[t] == -1)
            return flow;
        for (int i = 0; i < (int)G.size(); i++)
            h[i] += len[i];

        Flow f = limit;
        for (int v = t; v != s; v = prev[v])
            f = min(f, G[prev[v]][prev_num[v]].cap);
        for (int v = t; v != s; v = prev[v])
        {
            Edge &e = G[prev[v]][prev_num[v]];
            e.cap -= f;
            G[e.dst][e.rev].cap += f;
        }
        limit -= f;
        flow += f;
        cost += f * h[t];
    }
    return flow;
}

};

using MinCostMaxFlow = MIN_COST_MAX_FLOW::MinCostMaxFlow;

```

3.9 General Graph Matching

```

const int N = 514, E = (2e5) * 2;
struct Graph{
    int to[E], bro[E], head[N], e;
    int lnk[N], vis[N], stp, n;
    void init( int _n ){
        stp = 0; e = 1; n = _n;
        for( int i = 1 ; i <= n ; i ++ )
            lnk[i] = vis[i] = 0;
    }
}

```

```

void add_edge(int u,int v){
    to[e]=v,bro[e]=head[u],head[u]=e++;
    to[e]=u,bro[e]=head[v],head[v]=e++;
}
bool dfs(int x){
    vis[x]=stp;
    for(int i=head[x];i;i=bro[i]){
        int v=to[i];
        if(!lnk[v]){
            lnk[x]=v,lnk[v]=x;
            return true;
        }else if(vis[lnk[v]]<stp){
            int w=lnk[v];
            lnk[x]=v,lnk[v]=x,lnk[w]=0;
            if(dfs(w)){
                return true;
            }
            lnk[w]=v,lnk[v]=w,lnk[x]=0;
        }
    }
    return false;
}
int solve(){
    int ans = 0;
    for(int i=1;i<=n;i++){
        if(!lnk[i]){
            stp++; ans += dfs(i);
        }
    }
    return ans;
}
} graph;

```

4 Geometry

4.1 Smallest Circle - Emo Welzl (Contain all points)

```

#include <bits/stdc++.h>
using namespace std;
using ld = double;

```

```

typedef pair<ld, ld> point;
typedef pair<point, ld> circle;
#define X first
#define Y second

```

```

// O(n)
// Remember to change size of set points
// All point must save in array a[] below

```

```

namespace emowelzl
{
    const int N = 100005; // Size of set points
    int n;
    point a[N];

    point operator+(point a, point b)
    {
        return point(a.X + b.X, a.Y + b.Y);
    }
    point operator-(point a, point b) { return point(a.X - b.X, a.Y - b.Y); }
    point operator/(point a, ld x) { return point(a.X / x, a.Y / x); }
    ld abs(point a) { return sqrt(a.X * a.X + a.Y * a.Y); }

    point center_from(ld bx, ld by, ld cx, ld cy)
    {
        ld B = bx * bx + by * by, C = cx * cx + cy * cy, D = bx * cy - by * cx;
        return point((cy * B - by * C) / (2 * D), (bx * C - cx * B) / (2 * D));
    }

    circle circle_from(point A, point B, point C)
    {
        point I = center_from(B.X - A.X, B.Y - A.Y, C.X - A.X, C.Y - A.Y);
    }
}

```

```

        return circle(I + A, abs(I));
    }

    circle f(int n, vector<point> T)
    {
        if (T.size() == 3 || n == 0)
        {
            if (T.size() == 0)
                return circle(point(0, 0), -1);
            if (T.size() == 1)
                return circle(T[0], 0);
            if (T.size() == 2)
                return circle((T[0] + T[1]) / 2, abs(T[0] - T[1]) / 2);
            return circle_from(T[0], T[1], T[2]);
        }
        random_shuffle(a + 1, a + n + 1);
        circle Result = f(0, T);
        for (int i = 1; i <= n; i++)
            if (abs(Result.X - a[i]) > Result.Y + 1e-9)
            {
                T.push_back(a[i]);
                Result = f(i - 1, T);
                T.pop_back();
            }
        return Result;
    }
};

int main()
{
    cin >> emowelzl::n;

    for (int i = 1; i <= emowelzl::n; ++i)
        cin >> emowelzl::a[i].X >> emowelzl::a[i].Y;

    cout << emowelzl::f(emowelzl::n, {}).Y * 2;
}

```

4.2 Closest pair of points in set

// Find pair of points that have closest distance

```

#include <iostream>
#include <cstdio>
#include <algorithm>
#include <iomanip>
#include <cmath>
#include <vector>

```

```

using namespace std;

using ll = long long;
using ld = long double;

```

```

const int N = 5e4 + 2;
const ll Inf = 1e17;
#define sq(x) ((x) * (x))

```

```

struct Point
{
    ll x, y;
    int id;

    Point(const ll &x = 0, const ll &y = 0) : x(x), y(y) {}

    Point operator-(const Point &a) const
    {
        return Point(x - a.x, y - a.y);
    }
    ll len()
    {
        return x * x + y * y;
    }
};

```

```

namespace ClosestPoint

```

```

{
    int n, xa, ya;
    ll ans;
    Point a[N];

    ll Bruteforce(int l, int r)
    {
        for (; l < r; ++l)
            for (int i = l + 1; i <= r; ++i)
                if (ans > (a[l] - a[i]).len())
                {
                    ans = (a[l] - a[i]).len();
                    xa = a[l].id;
                    ya = a[i].id;
                }
        return ans;
    }

    void Brute(vector<int> &s)
    {
        sort(s.begin(), s.end(), [&](const int &x, const int &y)
            { return a[x].y < a[y].y; });
        for (int i = 0; i < s.size(); ++i)
            for (int j = i + 1; j < s.size() && sq(abs(a[s[i]].y - a[s[j]].y)) <= ans; ++j)
                if (ans > (a[s[i]] - a[s[j]]).len())
                {
                    ans = (a[s[i]] - a[s[j]]).len();
                    xa = a[s[i]].id;
                    ya = a[s[j]].id;
                }
    }

    void DAC(int l, int r)
    {
        if (r - l <= 3)
        {
            Bruteforce(l, r);
            return;
        }
        int mid = (l + r) / 2;

        DAC(l, mid);
        DAC(mid + 1, r);

        vector<int> s;
        for (int i = l; i <= r; ++i)
            if (sq(a[i].x - a[mid].x) <= ans)
                s.push_back(i);

        Brute(s);
    }

    void calc()
    {
        sort(a + 1, a + n + 1, [&](const Point &a, const Point &b)
            { return a.x < b.x || (a.x == b.x && a.y < b.y); });
        ans = Inf;

        DAC(1, n);

        if (xa > ya)
            swap(xa, ya);

        cout << xa << " " << ya << "\n";
        cout << fixed << setprecision(6) << sqrt((ld)ans);
    }
};

Point a[N];
int n;

void Read()
{
    cin >> n;
    for (int i = 1; i <= n; ++i)
    {
        cin >> a[i].x >> a[i].y;
    }
}

```

```

    a[i].id = i;
}

void Solve()
{
    ClosestPoint::n = n;
    for (int i = 1; i <= n; ++i)
        ClosestPoint::a[i] = a[i];

    ClosestPoint::calc();
}

int32_t main()
{
    ios_base::sync_with_stdio(0);
    cin.tie(0);
    cout.tie(0);
    Read();
    Solve();
}

```

4.3 Manhattan MST

// Idea is to reduce number of edges which are candidates to be in the MST
// Then apply Kruskal algorithm to find MST

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

constexpr int N = 2e5 + 2;
constexpr ll Inf = 1e17;

namespace manhattanMST
{
    /// disjoint set union
    struct dsu
    {
        int par[N];
        dsu()
        {
            memset(par, -1, sizeof par);
        }

        int findpar(int v)
        {
            return par[v] < 0 ? v : par[v] = findpar(par[v]);
        }

        bool Merge(int u, int v)
        {
            u = findpar(u);
            v = findpar(v);
            if (u == v)
                return false;

            if (par[u] < par[v])
                swap(u, v);

            par[v] += par[u];
            par[u] = v;

            return true;
        }
    };

    /// Fenwick Tree Min
    struct FenwickTreeMin
    {
        pair<ll, int> a[N];
        int n;

        FenwickTreeMin(int n = 0)
        {

```

```

            Assign(n);
        }

        void Assign(int n)
        {
            this->n = n;
            fill(a, a + n + 1, make_pair(Inf, -1));
        }

        void Update(int p, pair<ll, int> v)
        {
            for (; p <= n; p += p & -p)
                a[p] = min(a[p], v);
        }

        pair<ll, int> Get(int p)
        {
            pair<ll, int> ans((Inf, -1));

            for (; p; p -= p & -p)
                ans = min(ans, a[p]);
            return ans;
        }
    };

    struct Edge
    {
        int u, v;
        ll w;
        Edge(const int &u = 0, const int &v = 0, const ll &w = 0)
            : u(u), v(v), w(w) {}
        bool operator<(const Edge &a) const
        {
            return w < a.w;
        }
    };

    int n;
    ll x[N], y[N];
    vector<Edge> edges;

    ll dist(int i, int j)
    {
        return abs(x[i] - x[j]) + abs(y[i] - y[j]);
    }

#define Find(x, v) (lower_bound(x.begin(), x.end(), v) - x.begin() + 1)
    void createEdge(int a1, int a2, int b1, int b2, int c1, int c2)
    {
        vector<array<ll, 4>> v;
        vector<ll> s;

        for (int i = 1; i <= n; i++)
        {
            v.push_back({a1 * x[i] + a2 * y[i],
                        b1 * x[i] + b2 * y[i],
                        c1 * x[i] + c2 * y[i],
                        i});
            s.emplace_back(b1 * x[i] + b2 * y[i]);
        }
        sort(s.begin(), s.end());
        s.resize(unique(s.begin(), s.end()) - s.begin());
        sort(v.begin(), v.end());

        FenwickTreeMin f(n);

        for (auto [num1, num2, cost, idx] : v)
        {
            num2 = Find(s, num2);

            int res = f.Get(num2).second;
            if (res != -1)
                edges.emplace_back(res, idx, dist(res, idx));

            f.Update(num2, make_pair(cost, idx));
        }

```

```

        }

        void calc()
        {
            edges.clear();

            createEdge(1, -1, -1, 0, 1, 1); // R1
            createEdge(-1, 1, 0, -1, 1, 1); // R2
            createEdge(-1, -1, 0, 1, 1, -1); // R3
            createEdge(1, 1, -1, 0, 1, -1); // R4
            createEdge(-1, 1, 1, 0, -1, -1); // R5
            createEdge(1, -1, 0, 1, -1, -1); // R6
            createEdge(1, 1, 0, -1, -1, 1); // R7
            createEdge(-1, -1, 1, 0, -1, 1); // R8

            dsu f;

            sort(edges.begin(), edges.end());
            vector<pair<int, int>> res;
            ll ans(0);

            for (auto i : edges)
                if (f.Merge(i.u, i.v))
                {
                    ans += i.w;
                    res.emplace_back(i.u, i.v);
                }

            cout << ans << "\n";

            for (auto i : res)
                cout << i.first << " " << i.second << "\n";
        }
    };

    int32_t main()
    {
        ios_base::sync_with_stdio(0);
        cin.tie(0);
        cout.tie(0);

        cin >> manhattanMST::n;

        for (int i = 1; i <= manhattanMST::n; ++i)
            cin >> manhattanMST::x[i] >> manhattanMST::y[i];

        manhattanMST::calc();
    }
}

```

4.4 Geometry with real number

```

template <class T> int sgn(T x) { return (x > 0) - (x < 0); }
template <class T>
struct Point {
    typedef Point P;
    T x, y;
    explicit Point(T x=0, T y=0) : x(x), y(y) {}
    bool operator<(P p) const { return tie(x,y) < tie(p.x,p.y); }
    bool operator==(P p) const { return tie(x,y)==tie(p.x,p.y); }
    P operator+(P p) const { return P(x+p.x, y+p.y); }
    P operator-(P p) const { return P(x-p.x, y-p.y); }
    P operator*(T d) const { return P(x*d, y*d); }
    P operator/(T d) const { return P(x/d, y/d); }
    T dot(P p) const { return x*p.x + y*p.y; }
    T cross(P p) const { return x*p.y - y*p.x; }
    T cross(P a, P b) const { return (a-*this).cross(b-*this); }
    T dist2() const { return x*x + y*y; }
    double dist() const { return sqrt((double)dist2()); }
    // angle to x-axis in interval [-pi, pi]
    double angle() const { return atan2(y, x); }
    P unit() const { return *this/dist(); } // makes dist()=1
    P perp() const { return P(-y, x); } // rotates +90 degrees
    P normal() const { return perp().unit(); }
}

```



```

// returns point rotated 'a' radians ccw around the origin
P rotate(double a) const {
    return P(x*cos(a)-y*sin(a),x*sin(a)+y*cos(a)); }
friend ostream& operator<<(ostream& os, P p) {
    return os << "(" << p.x << ", " << p.y << ")"; }
};

/* On Segment: p is on segment from s to e */
template<class P> bool onSegment(P s, P e, P p) {
    return p.cross(s, e) == 0 && (s - p).dot(e - p) <= 0;
}

/* Segment Intersection */
template<class P> vector<P> segInter(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(d, b),
        oc = a.cross(b, c), od = a.cross(b, d);
    // Checks if intersection is single non-endpoint point.
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
        return {(a * ob - b * oa) / (ob - oa)};
    set<P> s;
    if (onSegment(c, d, a)) s.insert(a);
    if (onSegment(c, d, b)) s.insert(b);
    if (onSegment(a, b, c)) s.insert(c);
    if (onSegment(a, b, d)) s.insert(d);
    return {all(s)};
}

/**
 * Apply the linear transformation (translation, rotation and
 * scaling) which takes line p0-p1 to line q0-q1 to point r.
 */
typedef Point<double> P;
P linearTransformation(const P& p0, const P& p1,
    const P& q0, const P& q1, const P& r) {
    P dp = p1-p0, dq = q1-q0, num(dp.cross(dq), dp.dot(dq));
    return q0 + P((r-p0).cross(num), (r-p0).dot(num))/dp.dist2
        ();
}

/**
 * Computes the pair of points at which two circles intersect.
 */
typedef Point<double> P;
bool circleInter(P a, P b, double r1, double r2, pair<P, P>* out) {
    if (a == b) { assert(r1 != r2); return false; }
    P vec = b - a;
    double d2 = vec.dist2(), sum = r1+r2, dif = r1-r2,
        p = (d2 + r1*r1 - r2*r2)/(d2*2), h2 = r1*r1 - p*p*
        d2;
    if (sum*sum < d2 || dif*dif > d2) return false;
    P mid = a + vec*p, per = vec.perp() * sqrt(fmax(0, h2) /
        d2);
    *out = {mid + per, mid - per};
    return true;
}

/**
 * Finds the tangents of two circles, external if r2 > 0, internal
 * if r2 < 0.
 */
template<class P>
vector<pair<P, P>> tangents(P c1, double r1, P c2, double r2) {
    P d = c2 - c1;
    double dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr;
    if (d2 == 0 || h2 < 0) return {};
    vector<pair<P, P>> out;
    for (double sign : {-1, 1}) {
        P v = (d * dr + d.perp() * sqrt(h2) * sign) / d2;
        out.push_back({c1 + v * r1, c2 + v * r2});
    }
    if (h2 == 0) out.pop_back();
    return out;
}

/**
 * Returns the area of the intersection of a circle with a ccw
 * polygon.
 */

```

```

typedef Point<double> P;
#define arg(p, q) atan2(p.cross(q), p.dot(q))
double circlePoly(P c, double r, vector<P> ps) {
    auto tri = [&](P p, P q) {
        auto r2 = r * r / 2;
        P d = q - p;
        auto a = d.dot(p)/d.dist2(), b = (p.dist2()-r*r)/d
            .dist2();
        auto det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        auto s = max(0., -a-sqrt(det)), t = min(1., -a+
            sqrt(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = q + d * (t-1);
        return arg(p,u) * r2 + u.cross(v)/2 + arg(v,q) *
            r2;
    };
    auto sum = 0.0;
    rep(i,0,sz(ps))
        sum += tri(ps[i] - c, ps[(i + 1) % sz(ps)] - c);
    return sum;
}

/**
 * The circumcircle of a triangle is the circle intersecting all
 * three vertices.
 * ccRadius returns the radius of the circle going through points
 * A, B and C and ccCenter returns the center of the same
 * circle.
 */
typedef Point<double> P;
double ccRadius(const P& A, const P& B, const P& C) {
    return (B-A).dist()*(C-B).dist()*(A-C).dist()/
        abs((B-A).cross(C-A))/2;
}
P ccCenter(const P& A, const P& B, const P& C) {
    P b = C-A, c = B-A;
    return A + (b*c.dist2()-c*b.dist2()).perp()/b.cross(c)/2;
}

/**
 * Computes the minimum circle that encloses a set of points.
 */
typedef Point<double> P;
pair<P, double> mec(vector<P> ps) {
    shuffle(all(ps), mt19937(time(0)));
    P o = ps[0];
    double r = 0, EPS = 1 + 1e-8;
    rep(i,0,sz(ps)) if ((o - ps[i]).dist() > r * EPS) {
        o = ps[i], r = 0;
        rep(j,0,i) if ((o - ps[j]).dist() > r * EPS) {
            o = (ps[i] + ps[j]) / 2;
            r = (o - ps[i]).dist();
            rep(k,0,j) if ((o - ps[k]).dist() > r *
                EPS) {
                o = ccCenter(ps[i], ps[j], ps[k]);
                r = (o - ps[i]).dist();
            }
        }
    }
    return {o, r};
}

/**
 * Returns true if p lies within the polygon.
 * If strict is true, it returns false for points on the boundary.
 * The algorithm uses products in intermediate steps so watch out
 * for overflow.
 */
template<class P>
bool inPolygon(vector<P> &p, P a, bool strict = true) {
    int cnt = 0, n = sz(p);
    rep(i,0,n) {
        P q = p[(i + 1) % n];
        if (onSegment(p[i], q, a)) return !strict;
        //or: if (segDist(p[i], q, a) <= eps) return !
            strict;
        cnt ^= ((a.y<p[i].y) - (a.y<q.y)) * a.cross(p[i],
            q) > 0;
    }
}

```

```

    }
    return cnt;
}

/**
 * Returns the center of mass for a polygon.
 */
typedef Point<double> P;
P polygonCenter(const vector<P>& v) {
    P res(0, 0); double A = 0;
    for (int i = 0, j = sz(v) - 1; i < sz(v); j = i++) {
        res = res + (v[i] + v[j]) * v[j].cross(v[i]);
        A += v[j].cross(v[i]);
    }
    return res / A / 3;
}

/**
 * Returns a vector with the vertices of a polygon with everything
 * to the left of the vector going from s to e cut away.
 */
typedef Point<double> P;
vector<P> polygonCut(const vector<P>& poly, P s, P e) {
    vector<P> res;
    rep(i,0,sz(poly)) {
        P cur = poly[i], prev = i ? poly[i-1] : poly.back
            ();
        auto a = s.cross(e, cur), b = s.cross(e, prev);
        if ((a < 0) != (b < 0))
            res.push_back(cur + (prev - cur) * (a / (a
                - b)));
        if (a < 0)
            res.push_back(cur);
    }
    return res;
}

/**
 * Line-convex polygon intersection. The polygon must be ccw and
 * have no collinear points. lineHull(line, poly) returns a
 * pair describing the intersection of a line with the polygon:
 * * \begin{itemize}
 * * \item $(-1, -1)$ if no collision,
 * * \item $(i, -1)$ if touching the corner $i$,
 * * \item $(i, i)$ if along side $(i, i+1)$,
 * * \item $(i, j)$ if crossing sides $(i, i+1)$ and $(j, j+1)$.
 * * \end{itemize}
 */
#define cmp(i,j) sgn(dir.perp().cross(poly[(i)%n]-poly[(j)%n]))
#define extr(i) cmp(i + 1, i) >= 0 && cmp(i, i - 1 + n) < 0
template <class P> int extrVertex(vector<P>& poly, P dir) {
    int n = sz(poly), lo = 0, hi = n;
    if (extr(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (extr(m)) return m;
        int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi :
            lo) = m;
    }
    return lo;
}

#define cmpL(i) sgn(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
    int endA = extrVertex(poly, (a - b).perp());
    int endB = extrVertex(poly, (b - a).perp());
    if (cmpL(endA) < 0 || cmpL(endB) > 0)
        return {-1, -1};
    array<int, 2> res;
    rep(i,0,2) {
        int lo = endB, hi = endA, n = sz(poly);
        while ((lo + 1) % n != hi) {
            int m = ((lo + hi + (lo < hi ? 0 : n)) /
                2) % n;
            (cmpL(m) == cmpL(endB) ? lo : hi) = m;
        }
    }
}

```

```

        res[i] = (lo + !cmpL(hi)) % n;
        swap(endA, endB);
    }
    if (res[0] == res[1]) return {res[0], -1};
    if (!cmpL(res[0]) && !cmpL(res[1]))
        switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly)) {
            case 0: return {res[0], res[0]};
            case 2: return {res[1], res[1]};
        }
    return res;
}

```

5 Numerical algorithms

5.1 SQRT For Loop

```

// Calculate n/1 + n/2 + ... + n/n

#define Cal(a, b) ((b) - (a) + 1)

ll calc(ll n)
{
    ll ans = 0;

    for (ll i = 1; i <= n / i; ++i)
        ans += Cal(n / (i + 1) + 1, n / i) * i;

    for (ll i = 1; i < n / i; ++i)
        ans += n / i;

    return ans;
}

```

5.2 Rabin Miller - Prime Checker

// There is another version of Rabin Miller using random in the implementation of Pollard Rho

```

ll mul(ll a, ll b, ll mod)
{
    a %= mod;
    b %= mod;
    ll q = (ld)a * b / mod;
    ll r = a * b - q * mod;
    return (r % mod + mod) % mod;
}

ll pow(ll a, ll n, ll m)
{
    ll result = 1;
    a %= m;
    while (n > 0)
    {
        if (n & 1)
            result = mul(result, a, m);
        n >>= 1;
        a = mul(a, a, m);
    }
    return result;
}

pair<ll, ll> factor(ll n)
{
    ll s = 0;
    while ((n & 1) == 0)
    {
        s++;
        n >>= 1;
    }
    return {s, n};
}

bool test(ll s, ll d, ll n, ll witness)
{
    if (n == witness)
        return true;
    ll p = pow(witness, d, n);
    if (p == 1)

```

```

        return true;
    for (; s > 0; s--)
    {
        if (p == n - 1)
            return true;
        p = mul(p, p, n);
    }
    return false;
}

bool miller(ll n)
{
    if (n < 2)
        return false;
    if ((n & 1) == 0)
        return n == 2;
    ll s, d;
    tie(s, d) = factor(n - 1);
    return test(s, d, n, 2) && test(s, d, n, 3) && test(s, d, n,
        5) &&
        test(s, d, n, 7) && test(s, d, n, 11) && test(s, d, n,
        13) &&
        test(s, d, n, 17) && test(s, d, n, 19) && test(s, d, n,
        23);
}

```

5.3 Chinese Remain Theorem

```

pair<ll, ll> exteuclid(ll a, ll b){
    if(b == 0) return {1, 0};
    pair<ll, ll> q = gcd(b, a % b);
    return {q.second, q.first - q.second * (a / b)};
}

/**
 * x % m1 == x1
 * x % m2 == x2
 * => x % lcm(m1, m2) == res
 */
ll solve(ll x1, ll m1, ll x2, ll m2) {
    ll g = __gcd(m1, m2);
    if((x2 - x1) % g) return -1; // no sol
    m1 /= g; m2 /= g;
    pair<ll, ll> p = exteuclid(m1, m2);
    ll lcm = m1 * m2 * g;
    ll res = p.first * (x2 - x1) * m1 + x1;
    return (res % lcm + lcm) % lcm;
}

```

5.4 Pollard Rho - Factorialize

// You can change code Rabin-Miller (preposition)

```

struct PollardRho
{
    ll n;
    map<ll, int> ans;
    PollardRho(ll n) : n(n) {}
    ll random(ll u)
    {
        return abs(rand()) % u;
    }
    ll mul(ll a, ll b, ll mod)
    {
        a %= mod;
        b %= mod;
        ll q = (ld)a * b / mod;
        ll r = a * b - q * mod;
        return (r % mod + mod) % mod;
    }

    ll pow(ll a, ll b, ll m)
    {
        ll ans = 1;
        a %= m;
        for (; b; b >>= 1)
        {
            if (b & 1)

```

```

            ans = mul(ans, a, m);
            a = mul(a, a, m);
        }
        return ans;
    }

    pair<ll, ll> factor(ll n)
    {
        ll s = 0;
        while ((n & 1) == 0)
        {
            s++;
            n >>= 1;
        }
        return {s, n};
    }

    // Rabin - Miller
    bool miller(ll n)
    {
        if (n < 2)
            return 0;
        if (n == 2)
            return 1;
        ll s = 0, m = n - 1;
        while (m % 2 == 0)
        {
            s++;
            m >>= 1;
        }

        // 1 - 0.9 ^ 40
        for (int it = 1; it <= 40; it++)
        {
            ll u = random(n - 2) + 2;
            ll f = pow(u, m, n);
            if (f == 1 || f == n - 1)
                continue;
            for (int i = 1; i < s; i++)
            {
                f = mul(f, f, n);
                if (f == 1)
                    return 0;
                if (f == n - 1)
                    break;
            }
            if (f != n - 1)
                return 0;
        }
        return 1;
    }

    ll f(ll x, ll n)
    {
        return (mul(x, x, n) + 1) % n;
    }

    // Find a factor
    ll findfactor(ll n)
    {
        ll x = random(n - 1) + 2;
        ll y = x;
        ll p = 1;
        while (p == 1)
        {
            x = f(x, n);
            y = f(f(y, n), n);
            p = __gcd(abs(x - y), n);
        }
        return p;
    }

    // prime factorization
    void pollard_rho(ll n)
    {
        if (n <= 1000000)
        {
            for (int i = 2; i * i <= n; i++)
            {
                while (n % i == 0)
                {
                    ans[i]++;

```

```

        n /= i;
    }
    if (n > 1)
        ans[n]++;
    return;
}

if (miller(n))
{
    ans[n]++;
    return;
}

ll p = 0;

while (p == 0 || p == n)
    p = findfactor(n);

pollard_rho(n / p);
pollard_rho(p);
};

```

5.5 FFT

```

using cd = complex<double>;
const double PI = acos(-1);
// invert == true means Interpolation
// invert == false means dft
void fft(vector<cd> &a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1)
            j ^= bit;
        j ^= bit;
        if (i < j)
            swap(a[i], a[j]);
    }

    for (int len = 2; len <= n; len <= 1) {
        double ang = 2 * PI / len * (invert ? -1 : 1);
        cd wlen(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            cd w(1);
            for (int j = 0; j < len / 2; j++) {
                cd u = a[i + j],
                    v = a[i + j + len / 2] * w;
                a[i + j] = u + v;
                a[i + j + len / 2] = u - v;
                w *= wlen;
            }
        }

        if (invert) {
            for (cd &x : a)
                x /= n;
        }
    }
}

vector<long long> multiply(vector<long long> &A, vector<long long>
    &B) {
    vector<cd> fa(A.begin(), A.end()), fb(B.begin(), B.end());

    int n1 = (int)A.size(), n2 = (int)B.size();
    int n = 1;
    while (n < n1 + n2 - 1) n <= 1;

    fa.resize(n);
    fb.resize(n);

    fft(fa, false);
    fft(fb, false);
    for (int i = 0; i < n; i++) fa[i] *= fb[i];
    fft(fa, true);

```

```

vector<long long> res(n1 + n2 - 1);
for (int i = 0; i < (int)res.size(); i++) {
    res[i] = (long long) llround(fa[i].real()); // round to
        nearest integer
}
return res;
}

```

5.6 FFT (Mod 998244353)

```

// NTT (FFT mod 998244353) + convolution
static const int MOD = 998244353;
static const int G = 3;

long long modpow(long long a, long long e) {
    long long r = 1 % MOD;
    a %= MOD;
    while (e > 0) {
        if (e & 1) r = r * a % MOD;
        a = a * a % MOD;
        e >>= 1;
    }
    return r;
}

// invert=false: forward NTT, invert=true: inverse NTT
void fft_mod(vector<int> &a, bool invert) {
    int n = (int)a.size();
    static vector<int> rev;
    static vector<int> roots(0, 1);

    // bit-reversal permutation
    if ((int)rev.size() != n) {
        int k = __builtin_ctz(n);
        rev.assign(n, 0);
        for (int i = 0; i < n; i++) {
            rev[i] = 0;
            for (int j = 0; j < k; j++)
                if (i & (1 << j)) rev[i] |= 1 << (k - 1 - j);
        }
    }

    // roots of unity
    if ((int)roots.size() < n) {
        int k = __builtin_ctz((int)roots.size());
        roots.resize(n);
        while ((1 << k) < n) {
            long long e = modpow(G, (MOD - 1) >> (k + 1));
            for (int i = 1 << (k - 1); i < (1 << k); i++) {
                roots[2 * i] = roots[i];
                roots[2 * i + 1] = (long long)roots[i] * e % MOD;
            }
            k++;
        }
    }

    for (int i = 0; i < n; i++)
        if (i < rev[i]) swap(a[i], a[rev[i]]);

    for (int len = 1; len < n; len <= 1) {
        for (int i = 0; i < n; i += 2 * len) {
            for (int j = 0; j < len; j++) {
                int u = a[i + j];
                int v = (long long)a[i + j + len] * roots[len + j]
                    % MOD;
                int x = u + v;
                if (x >= MOD) x -= MOD;
                a[i + j] = x;
                x = u - v;
                if (x < 0) x += MOD;
                a[i + j + len] = x;
            }
        }
    }

    if (invert) {

```

```

// inverse via reversing (valid for this root table
    construction)
reverse(a.begin() + 1, a.end());
long long inv_n = modpow(n, MOD - 2);
for (int &x : a) x = (long long)x * inv_n % MOD;
}

vector<long long> multiply(const vector<long long> &A, const
    vector<long long> &B) {
    if (A.empty() || B.empty()) return {};
    int n1 = (int)A.size(), n2 = (int)B.size();
    int n = 1;
    while (n < n1 + n2 - 1) n <= 1;

    vector<int> fa(n, 0), fb(n, 0);
    for (int i = 0; i < n1; i++) fa[i] = (int)(A[i] % MOD);
    for (int i = 0; i < n2; i++) fb[i] = (int)(B[i] % MOD);

    fft_mod(fa, false);
    fft_mod(fb, false);
    for (int i = 0; i < n; i++) fa[i] = (long long)fa[i] * fb[i] %
        MOD;
    fft_mod(fa, true);

    vector<long long> res(n1 + n2 - 1);
    for (int i = 0; i < (int)res.size(); i++) res[i] = fa[i];
    return res;
}

```

5.7 FFT Mod (Template by Tran Khoi Nguyen)

```

struct FFTmod
{
    typedef complex<double> C;
    const ll M = mod;
    void fft(vector<C> &a)
    {
        int n = a.size(), L = 31 - __builtin_clz(n);
        static vector<complex<long double>> R(2, 1);
        static vector<C> rt(2, 1); // (^ 10% faster if double)
        for (static int k = 2; k < n; k *= 2)
        {
            R.resize(n);
            rt.resize(n);
            auto x = polar(1.0L, acos(-1.0L) / k);
            for (int i = k; i < 2 * k; i++)
                rt[i] = R[i] = i & 1 ? R[i / 2] * x : R[i / 2];
        }
        vector<int> rev(n);
        for (int i = 0; i < n; i++)
            rev[i] = (rev[i / 2] | (i & 1) << L) / 2;

        for (int i = 0; i < n; i++)
            if (i < rev[i])
                swap(a[i], a[rev[i]]);
        for (int k = 1; k < n; k *= 2)
            for (int i = 0; i < n; i += 2 * k)
                for (int j = 0; j < k; j++)
                {
                    // C z = rt[j+k] * a[i+j+k]; // (25% faster if
                        hand-rolled) // include-line
                    auto x = (double *)&rt[j + k], y = (double *)&
                        a[i + j + k]; // exclude-line
                    C z(x[0] * y[0] - x[1] * y[1], x[0] * y[1] + x
                        [1] * y[0]); // exclude-line
                    a[i + j + k] = a[i + j] - z;
                    a[i + j] += z;
                }
        }

    typedef vector<ll> vl;
    vl convMod(const vl &a, const vl &b)
    {
        vl res((int)a.size() + b.size() - 1);

```

```

int B = 32 - __builtin_clz(res.size()), n = 1 << B, cut =
    int(sqrt(M));

vector<C> L(n), R(n), outs(n), outl(n);

for (int i = 0; i < (int)a.size(); i++)
    L[i] = C((int)a[i] / cut, (int)a[i] % cut);

for (int i = 0; i < (int)b.size(); i++)
    R[i] = C((int)b[i] / cut, (int)b[i] % cut);

fft(L), fft(R);

for (int i = 0; i < n; i++)
{
    int j = -i & (n - 1);
    outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
    outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / 1i;
}

fft(outl), fft(outs);

for (int i = 0; i < (int)res.size(); i++)
{
    ll av = (ll)(real(outl[i]) + .5);
    ll cv = (ll)(imag(outs[i]) + .5);
    ll bv = (ll)(imag(outl[i]) + .5) + (ll)(real(outs[i])
        + .5);
    res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
}
return res;
};

```

5.8 Fast Walsh-Hadamard Transform (FWHT)

```

void FST(vector<int> &a, bool inv) {
    for (int n = (int) a.size(), step = 1; step < n; step *= 2)
        for (int i = 0; i < n; i += 2 * step)
            for (int j = 0; j < step; ++j) {
                int u = a[i + j];
                int v = a[i + j + step];

                // tie(a[i + j], a[i + j + step]) = (inv ?
                // make_pair(v, u - v) : make_pair(u + v, v));
                // AND
                // tie(a[i + j], a[i + j + step]) = (inv ?
                // make_pair(u - v, v) : make_pair(u, u + v));
                // OR
                tie(a[i + j], a[i + j + step]) = make_pair(u + v,
                    u - v); // XOR
            }

    if (inv) {
        for (int &x : a) x /= (int) a.size(); // XOR only
    }
}

vector<int> conv(vector<int> &a, vector<int> &b) {
    int n = 1, need = (int)a.size() + (int)b.size() - 1;
    while (n < need) n <<= 1;
    a.resize(n, 0); b.resize(n, 0);

    FST(a, 0); FST(b, 0);
    for (int i = 0; i < (int) a.size(); ++i) a[i] *= b[i];
    FST(a, 1);

    return a;
}

```

5.9 Fast Walsh-Hadamard Transform (FWHT Base m)

```

// m | MOD - 1 is required
int primitive_mth_root(int m, int mod) {
    int g = PrimitiveRoot(mod);
    return Pow(g, (mod - 1) / m, mod);
}

```

```

}

void fwht_m(vector<ll>& a, int m, int omega, bool inverse) {
    const int SZ = (int)a.size();
    ll w = inverse ? Pow(omega, m-1, mod) : omega;

    vector<ll> wp(m, 1);
    for (int t = 1; t < m; ++t) wp[t] = wp[t-1] * w % mod;

    vector<ll> Mmt(m * m);
    for (int k = 0; k < m; ++k)
        for (int t = 0; t < m; ++t)
            Mmt[k * m + t] = wp[(k * t) % m];

    int stages = 0;
    for (int len=1; len<SZ; len *= m) {
        ++stages;
        for (int base = 0; base < SZ; base += m * len) {
            for (int j = 0; j < len; ++j) {
                static vector<ll> x, y;
                x.assign(m, 0); y.assign(m, 0);
                for (int t=0; t<m; ++t) x[t] = a[base + j + t *
                    len];

                for (int k = 0; k < m; ++k) {
                    ll s = 0;
                    const ll* row = &Mmt[k*m];
                    for (int t = 0; t < m; ++t) {
                        s += x[t] * row[t];
                        s %= mod;
                    }
                    y[k] = s;
                }
                for (int k = 0; k < m; ++k) a[base + j + k * len]
                    = y[k];
            }
        }
    }

    if (inverse) {
        ll invm = mod_inv(m); ll scale = 1;
        for (int s = 0; s < stages; ++s) scale = scale * invm %
            mod;
        for (ll &v : a) v = v * scale % mod;
    }
}

```

5.10 Count Primes

```

// To initialize, call init_count_primes() first.
// Function count_primes(n) will compute the number of
// prime numbers lower than or equal to n.
//
// Time complexity: Around O(n ^ 0.75)

constexpr int N = 1e5 + 5; // keep N larger than max(sqrt(n) + 2)
bool prime[N];
int prec[N];
vector<int> P;

ll rec(ll n, int k)
{
    if (n <= 1 || k < 0)
        return 0;
    if (n <= P[k])
        return n - 1;

    if (n < N && 11(P[k]) * P[k] > n)
        return n - 1 - prec[n] + prec[P[k]];

    const int LIM = 250;
    static int memo[LIM * LIM][LIM];
    bool ok = n < LIM * LIM;
    if (ok && memo[n][k])
        return memo[n][k];

    ll ret = n / P[k] - rec(n / P[k], k - 1) + rec(n, k - 1);
}

```

```

    if (ok)
        memo[n][k] = ret;
    return ret;
}

void init_count_primes()
{
    prime[2] = true;
    for (int i = 3; i < N; i += 2)
        prime[i] = true;
    for (int i = 3, j; i * i < N; i += 2)
        if (prime[i])
            for (j = i * i; j < N; j += i + i)
                prime[j] = false;

    for (int i = 1; i < N; ++i)
        if (prime[i])
            P.push_back(i);

    for (int i = 1; i < N; ++i)
        prec[i] = prec[i - 1] + prime[i];
}

ll count_primes(ll n)
{
    if (n < N)
        return prec[n];
    int k = prec[(int)sqrt(n) + 1];
    return n - 1 - rec(n, k) + prec[P[k]];
}

```

5.11 Interpolation (Mod a prime)

```

// You can change mod into other prime number
// update k to the degree of polynomial
// Just work when we know a[1] = P(1), a[2] = P(2), ..., a[k] = P(k)
// [The degree of P(x) is k-1]
// update() then build() then cal()

/*
 * Complexity: O(Nlog(mod), N)
 */

```

```

constexpr ll mod = 1e9 + 7; // Change mod here
constexpr ll N = 1e5 + 5; // Change size here

```

struct Interpolation

```

{
    ll a[N], fac[N], ifac[N], prf[N], suf[N];
    int k;

    ll Pow(ll a, ll b)
    {
        ll ans(1);
        for (; b; b >=> 1)
        {
            if (b & 1)
                ans = ans * a % mod;
            a = a * a % mod;
        }

        return ans;
    }

    void upd(int u, ll v)
    {
        a[u] = v;
    }

    void build()
    {
        fac[0] = ifac[0] = 1;
        for (int i = 1; i < N; i++)
        {
            fac[i] = (long long)fac[i - 1] * i % mod;
            ifac[i] = Pow(fac[i], mod - 2);
        }
    }
}

```

```

}

// Calculate P(x)
ll calc(int x)
{
    prf[0] = suf[k + 1] = 1;

    for (int i = 1; i <= k; i++)
        prf[i] = prf[i - 1] * (x - i + mod) % mod;

    for (int i = k; i >= 1; i--)
        suf[i] = suf[i + 1] * (x - i + mod) % mod;

    ll res = 0;

    for (int i = 1; i <= k; i++)
    {
        if (!(k - i) & 1)
            res = (res + prf[i - 1] * suf[i + 1] % mod * ifac[
                i - 1] % mod * ifac[k - i] % mod * a[i]) %
                mod;
        else
            res = (res - prf[i - 1] * suf[i + 1] % mod * ifac[
                i - 1] % mod * ifac[k - i] % mod * a[i] % mod
                + mod) % mod;
    }

    return res;
}
};

```

5.12 Bignum

/// M is the number of digits in the answer
/// In case that we don't use multiplication, let BASE be 1e17 or 1e18
/// a = Bignum("5")
/// The operator / is only for integer, the result is integer too

```

using cd = complex<long double>;
const long double PI = acos(-1);
const int M = 2000;
const ll BASE = 1e8;
const int gd = log10(BASE);
const int maxn = M / gd + 1;
struct Bignum
{
    int n;
    ll a[maxn];
    Bignum(ll x = 0)
    {
        memset(a, 0, sizeof a);
        n = 0;
        do
        {
            a[n++] = x % BASE;
            x /= BASE;
        } while (x);
    }
    Bignum(const string &s)
    {
        Convert(s);
    }
    ll stoll(const string &s)
    {
        ll ans(0);
        for (auto i : s)
            ans = ans * 10 + i - '0';
        return ans;
    }
    void Convert(const string &s)
    {
        memset(a, 0, sizeof a);
        n = 0;
        for (int i = s.size() - 1; ~i; --i)
        {
            int j = max(0, i - gd + 1);

```

```

            a[n++] = stoll(s.substr(j, i - j + 1));
            i = j;
        }
        fix();
    }
    void fix()
    {
        ++n;
        for (int i = 0; i < n - 1; ++i)
        {
            a[i + 1] += a[i] / BASE;
            a[i] %= BASE;
            if (a[i] < 0)
            {
                a[i] += BASE;
                --a[i + 1];
            }
        }
        while (n > 1 && a[n - 1] == 0)
            --n;
    }
    Bignum &operator+=(const Bignum &x)
    {
        n = max(n, x.n);
        for (int i = 0; i < n; ++i)
            a[i] += x.a[i];
        fix();
        return *this;
    }
    Bignum &operator-=(const Bignum &x)
    {
        for (int i = 0; i < x.n; ++i)
            a[i] -= x.a[i];
        fix();
        return *this;
    }
    Bignum &operator*=(const Bignum &x)
    {
        vector<ll> c(x.n + n, 0);
        for (int i = 0; i < n; ++i)
            for (int j = 0; j < x.n; ++j)
                c[i + j] += a[i] * x.a[j];
        n += x.n;
        for (int i = 0; i < n; ++i)
            a[i] = c[i];
        fix();
        return *this;
    }
    Bignum &operator/=(const ll &x)
    {
        ll r = 0;
        for (int i = n - 1; i > -1; --i)
        {
            r = r * BASE + a[i];
            a[i] = r / x;
            r %= x;
        }
        fix();
        return *this;
    }
    Bignum operator+(const Bignum &s)
    {
        Bignum c;
        copy(a, a + n, c.a);
        c.n = n;
        c += s;
        return c;
    }
    Bignum operator-(const Bignum &s)
    {
        Bignum c;
        copy(a, a + n, c.a);
        c.n = n;
        c -= s;
        return c;
    }
    Bignum operator*(const Bignum &s)
    {

```

```

        Bignum c;
        copy(a, a + n, c.a);
        c.n = n;
        c *= s;
        return c;
    }
    Bignum operator/(const ll &x)
    {
        Bignum c;
        copy(a, a + n, c.a);
        c.n = n;
        c /= x;
        return c;
    }
    ll operator%(const ll &x)
    {
        ll ans(0);
        for (int i = n - 1; ~i; --i)
            ans = (ans * BASE + a[i]) % x;
        return ans;
    }
    int com(const Bignum &s) const
    {
        if (n < s.n)
            return 1;
        if (n > s.n)
            return 2;
        for (int i = n - 1; i > -1; --i)
            if (a[i] > s.a[i])
                return 2;
            else if (a[i] < s.a[i])
                return 1;
        return 3;
    }
    bool operator<(const Bignum &s) const
    {
        return com(s) == 1;
    }
    bool operator>(const Bignum &s) const
    {
        return com(s) == 2;
    }
    bool operator==(const Bignum &s) const
    {
        return com(s) == 3;
    }
    bool operator<=(const Bignum &s) const
    {
        return com(s) != 2;
    }
    bool operator>=(const Bignum &s) const
    {
        return com(s) != 1;
    }
    void read()
    {
        string s;
        cin >> s;
        Convert(s);
    }
    void print()
    {
        int i = n;
        while (i > 0 && a[i] == 0)
            --i;
        cout << a[i];
        for (--i; ~i; --i)
            cout << setw(gd) << setfill('0')
                << a[i];
    }
};

```

5.13 Bignum with FFT multiplication

*/// Replace function *= in Bignum implementation with below code:*
void fft(vector<cd> &a, bool invert)
{

```

int n = a.size();
for (int i = 1, j = 0; i < n; i++)
{
    int bit = n >> 1;
    for (; j & bit; bit >>= 1)
        j ^= bit;
    j ^= bit;
    if (i < j)
        swap(a[i], a[j]);
}
for (int len = 2; len <= n; len <=<= 1)
{
    double ang = 2 * PI / len * (invert ? -1 : 1);
    cd wlen(cos(ang), sin(ang));
    for (int i = 0; i < n; i += len)
    {
        cd w(1);
        for (int j = 0; j < len / 2; j++)
        {
            cd u = a[i + j], v = a[i + j + len / 2] * w;
            a[i + j] = u + v;
            a[i + j + len / 2] = u - v;
            w *= wlen;
        }
    }
    if (invert)
    {
        for (cd &x : a)
            x /= n;
    }
}
Bignum &operator*(const Bignum &x)
{
    int m = 1;
    while (m < n + x.n)
        m <=<= 1;
    vector<cd> fa(m), fb(m);
    for (int i = 0; i < m; ++i)
    {
        fa[i] = a[i];
        fb[i] = x.a[i];
    }
    fft(fa, false); /// dft
    fft(fb, false); /// dft
    for (int i = 0; i < m; i++)
        fa[i] *= fb[i];
    fft(fa, true); /// Interpolation
    n = m;
    for (int i = 0; i < n; ++i)
        a[i] = round(fa[i].real());
    fix();
    return *this;
}

```

5.14 Tonelli Shanks (Find square root modulo prime)

/*
Takes as input an odd prime p and $n < p$ and returns r such that $r^2 \equiv n \pmod{p}$.
There's exist r if and only if $n^{(p-1)/2} \equiv 1 \pmod{p}$
*/

using ll = int; // Change type of data here

```

ll Pow(ll a, ll b, ll mod)
{
    ll ans(1);

    for (; b; b >>= 1)
    {
        if (b & 1)
            ans = ans * a % mod;
        a = a * a % mod;
    }

    return ans;
}

```

```

}

ll tonelli_shanks(ll n, ll p)
{
    ll s = 0;
    ll q = p - 1;
    while ((q & 1) == 0)
    {
        q /= 2;
        ++s;
    }
    if (s == 1)
    {
        ll r = Pow(n, (p + 1) / 4, p);
        if ((r * r) % p == n)
            return r;
        return 0;
    }
    // Find the first quadratic non-residue z by brute-force search
    ll z = 1;
    while (Pow(++z, (p - 1) / 2, p) != p - 1)
        ;
    ll c = Pow(z, q, p);
    ll r = Pow(n, (q + 1) / 2, p);
    ll t = Pow(n, q, p);
    ll m = s;
    while (t != 1)
    {
        ll tt = t;
        ll i = 0;
        while (tt != 1)
        {
            tt = (tt * tt) % p;
            ++i;
            if (i == m)
                return 0;
        }
        ll b = Pow(c, Pow(2, m - i - 1, p - 1), p);
        ll b2 = (b * b) % p;
        r = (r * b) % p;
        t = (t * b2) % p;
        c = b2;
        m = i;
    }
    if ((r * r) % p == n)
        return r;
    return -1; // Can't find
}

```

5.15 Discrete Logarithm (Find x that $a^x \equiv b \pmod{m}$)

/// Returns minimum x for which $a^x \equiv b \pmod{m}$.
/// Returns -1 if x isn't exist

using ll = int;

```

ll DiscreteLogarithm(ll a, ll b, ll m)
{
    a %= m, b %= m;
    ll k = 1, add = 0, g;
    while ((g = __gcd(a, m)) > 1)
    {
        if (b == k)
            return add;
        if (b % g)
            return -1;
        b /= g, m /= g, ++add;
        k = (k * 111 * a / g) % m;
    }

    ll n = sqrt((1d)m) + 1;
    ll an = 1;
    for (ll i = 0; i < n; ++i)
        an = (an * 111 * a) % m;
}

```

```

unordered_map<ll, ll> vals;
for (ll q = 0, cur = b; q <= n; ++q)
{
    vals[cur] = q;
    cur = (cur * 111 * a) % m;
}

for (ll p = 1, cur = k; p <= n; ++p)
{
    cur = (cur * 111 * an) % m;
    if (vals.count(cur))
    {
        ll ans = n * p - vals[cur] + add;
        return ans;
    }
}
return -1;
}

```

5.16 Primitive Root (Exist k that $g^k \equiv a \pmod{n}$ for all a)

/*
 g is a primitive root modulo n if and only if for any integer a such that $\gcd(a, n) = 1$, there exists an integer k such that:
 $g^k \equiv a \pmod{n}$.

Primitive root modulo n exists if and only if:

- n is 1, 2, 4
- n is power of an odd prime number ($n = p^k$)
- n is twice power of an odd prime number ($n = 2 * p^k$)

This theorem was proved by Gauss in 1801.

*/

using ll = int; // Change type of data here

```

ll Pow(ll a, ll b, ll mod)
{
    ll ans(1);

    for (; b; b >>= 1)
    {
        if (b & 1)
            ans = ans * a % mod;
        a = a * a % mod;
    }

    return ans;
}

ll GetPhi(ll n)
{
    ll ans(1);

    for (ll i = 2; i * i <= n; ++i)
        if (n % i == 0)
        {
            while (n % i == 0)
            {
                n /= i;
                ans *= i;
            }

            ans = ans / i * (i - 1);
        }

    if (n != 1)
        ans *= n - 1;

    return ans;
}

```

```

ll PrimitiveRoot(ll p)
{
    vector<ll> fact;
    ll phi = GetPhi(p);
    ll n = phi;
}

```

```

for (int i = 2; i * i <= n; ++i)
    if (n % i == 0)
    {
        fact.push_back(i);
        while (n % i == 0)
            n /= i;
    }

if (n > 1)
    fact.push_back(n);

for (ll res = 2; res <= p; ++res)
{
    bool ok = true;

    for (int i = 0; i < fact.size() && ok; ++i)
        ok &= Pow(res, phi / fact[i], p) != 1;

    if (ok)
        return res;
}

return -1; // can't find

```

5.17 Discrete Root (Find x that $x^k \equiv a \pmod{n}$, n is a prime)

/*
Given a prime n and two integers a and k , find all x for which:
- $x^k \equiv a \pmod{n}$

Notice:
- In case $k = 2$, let's use Tonelli - Shanks
- You must insert my implementation of Discrete Logarithm and Primitive Root to run algorithm

```

*/

ll Pow(ll a, ll b, ll mod)
{
    ll ans(1);

    for (; b; b >>= 1)
    {
        if (b & 1)
            ans = ans * a % mod;
        a = a * a % mod;
    }

    return ans;
}

ll DiscreteRoot(ll a, ll k, ll n)
{
    ll g = PrimitiveRoot(n);
    ll v = Pow(g, k, n);
    ll ans = DiscreteLogarithm(v, a, n);

    if (ans == -1)
        return -1; // Can't find

    return Pow(g, ans, n);
}

```

5.18 Super Sieve of Primes (Code by RR)

```

// Sieve up to  $10^9$  by RR
namespace Sieve
{
    const int MAX = 1000000000LL;
    const int WHEEL = 3 * 5 * 7 * 11 * 13;
    const int N_SMALL_PRIMES = 6536; // cnt primes less than
        2^16
    const int SIEVE_SPAN = WHEEL * 64; // one iteration of
        segmented sieve
    const int SIEVE_SIZE = SIEVE_SPAN / 128 + 1;

```

```

uint64_t ONES[64]; // ONES[i] = 1<<i
int small_primes[N_SMALL_PRIMES]; // primes less than 2^16

// each element of sieve is a 64-bit bitmask.
// Each bit (0/1) stores whether the corresponding element is
// a prime number.
// We only need to store odd numbers
// -> 1st bitmask stores 3, 5, 7, 9, ...
uint64_t si[SIEVE_SIZE];
// for each 'wheel', we store the sieve pattern (i.e. what
// numbers cannot be primes)
uint64_t pattern[WHEEL];

inline void mark(uint64_t *s, int o) { s[o >> 6] |= ONES[o &
63]; }

inline int test(uint64_t *s, int o) { return (s[o >> 6] & ONES
[o & 63]) == 0; }

// update_sieve {{{
void update_sieve(int offset)
{
    // copy each wheel pattern to sieve
    for (int i = 0, k; i < SIEVE_SIZE; i += k)
    {
        k = std::min(WHEEL, SIEVE_SIZE - i);
        memcpy(si + i, pattern, sizeof(*pattern) * k);
    }

    // Correctly mark 1, 3, 5, 7, 11, 13 as not prime / primes
    if (offset == 0)
    {
        si[0] |= ONES[0];
        si[0] &= ~(ONES[1] | ONES[2] | ONES[3] | ONES[5] |
ONES[6]);
    }

    // sieve for primes >= 17 (stored in 'small_primes')
    for (int i = 0; i < N_SMALL_PRIMES; ++i)
    {
        int j = small_primes[i] * small_primes[i];
        if (j > offset + SIEVE_SPAN - 1)
            break;
        if (j > offset)
            j = (j - offset) >> 1;
        else
        {
            j = small_primes[i] - offset % small_primes[i];
            if ((j & 1) == 0)
                j += small_primes[i];
            j >>= 1;
        }
        while (j < SIEVE_SPAN / 2)
        {
            mark(si, j);
            j += small_primes[i];
        }
    }
}

// }}}

void sieve()
{
    // init small primes {{{
    for (int i = 0; i < 64; ++i)
        ONES[i] = 1ULL << i;

    // sieve to find small primes
    for (int i = 3; i < 256; i += 2)
    {
        if (test(si, i >> 1))
        {
            for (int j = i * i / 2; j < 32768; j += i)
                mark(si, j);
        }
    }

    // store primes >= 17 in 'small_primes' (we will sieve
    differently

```

```

// for primes 2, 3, 5, 7, 11, 13)
{
    int m = 0;
    for (int i = 8; i < 32768; ++i)
    {
        if (test(si, i))
            small_primes[m++] = i * 2 + 1;
    }
}

// For primes 3, 5, 7, 11, 13: we initialize wheel pattern
...
for (int i = 1; i < WHEEL * 64; i += 3)
    mark(pattern, i);
for (int i = 2; i < WHEEL * 64; i += 5)
    mark(pattern, i);
for (int i = 3; i < WHEEL * 64; i += 7)
    mark(pattern, i);
for (int i = 5; i < WHEEL * 64; i += 11)
    mark(pattern, i);
for (int i = 6; i < WHEEL * 64; i += 13)
    mark(pattern, i);

// Segmented sieve
long long sum_primes = 2;

for (int offset = 0; offset < MAX; offset += SIEVE_SPAN)
{
    update_sieve(offset);

    for (uint32_t j = 0; j < SIEVE_SIZE; j++)
    {
        uint64_t x = ~si[j];
        while (x)
        {
            uint32_t p = offset + (j << 7) + (
__builtin_ctzll(x) << 1) + 1;
            if (p > offset + SIEVE_SPAN - 1)
                break;
            if (p <= MAX)
            {
                // p is a prime
                x ^= (-x & x);
            }
        }
    }
}
};

```

6 Graph algorithms

6.1 Twosat (2-SAT)

```

// start from 0
// pos(V) is the vertex that represent V in graph
// neg(V) is the vertex that represent !V
// pos(V) ^ neg(V) = 1, use two functions below
// (U v V) <=> (!U -> V) <=> (!V -> U)
// You need do addEdge(represent(U), represent(V))
// solve() == false mean no answer
// Want to get the answer ?
// color[pos(U)] = 1 means we choose U
// otherwise, we don't
constexpr int N = 1e5 + 5; // Keep N double of n
inline int pos(int u) { return u << 1; }
inline int neg(int u) { return u << 1 | 1; }
struct TwoSAT
{
    int n, numComp, cntTarjan;
    vector<int> adj[N], stTarjan;
    int low[N], num[N], root[N], color[N];
    TwoSAT(int n) : n(n * 2)
    {
        memset(root, -1, sizeof root);
        memset(low, -1, sizeof low);

```



```

memset(num, -1, sizeof num);
memset(color, -1, sizeof color);
cntTarjan = 0;
stTarjan.clear();
}
void addEdge(int u, int v)
{
    adj[u ^ 1].push_back(v);
    adj[v ^ 1].push_back(u);
}
void tarjan(int u)
{
    stTarjan.push_back(u);
    num[u] = low[u] = cntTarjan++;
    for (int v : adj[u])
    {
        if (root[v] != -1)
            continue;
        if (low[v] == -1)
            tarjan(v);
        low[u] = min(low[u], low[v]);
    }
    if (low[u] == num[u])
    {
        while (1)
        {
            int v = stTarjan.back();
            stTarjan.pop_back();
            root[v] = numComp;
            if (u == v)
                break;
        }
        numComp++;
    }
}
bool solve()
{
    for (int i = 0; i < n; i++)
        if (root[i] == -1)
            tarjan(i);
    for (int i = 0; i < n; i += 2)
    {
        if (root[i] == root[i ^ 1])
            return 0;
        color[i] = (root[i] < root[i ^ 1]);
    }
    return 1;
}
};

```

6.2 Eulerian Path

```

// Path that goes all edges
// Start from 1
struct EulerianGraph
{
    vector<vector<pair<int, int>>> a;
    int num_edges;

    EulerianGraph(int n)
    {
        a.resize(n + 1);
        num_edges = 0;
    }

    void add_edge(int u, int v, bool undirected = true)
    {
        a[u].push_back(make_pair(v, num_edges));
        if (undirected)
            a[v].push_back(make_pair(u, num_edges));
        num_edges++;
    }

    vector<int> get_eulerian_path()
    {
        vector<int> path, s;
        vector<bool> was(num_edges);
    }
}

```

```

s.push_back(1);
// start of eulerian path
// directed graph: deg_out - deg_in == 1
// undirected graph: odd degree
// for eulerian cycle: any vertex is OK

while (!s.empty())
{
    int u = s.back();
    bool found = false;
    while (!a[u].empty())
    {
        int v = a[u].back().first;
        int e = a[u].back().second;
        a[u].pop_back();
        if (was[e])
            continue;
        was[e] = true;
        s.push_back(v);
        found = true;
        break;
    }
    if (!found)
    {
        path.push_back(u);
        s.pop_back();
    }
}
reverse(path.begin(), path.end());
return path;
};

```

6.3 Biconnected Component Tree

```

// Biconnected Component Tree
// 1 is the root of Tree
// n + i is the node that represent i-th bcc, its depth is even

const int N = 3e5 + 5; // Change size to n + number of bcc (For
                        // safety, set N >= 2 * n)
int n, nBicon, nTime;
int low[N], num[N];
vector<int> adj[N], nadj[N];
vector<int> s;

void dfs(int v, int p = -1)
{
    low[v] = num[v] = ++nTime;
    s.emplace_back(v);

    for (auto i : adj[v])
        if (i != p)
        {
            if (!num[i])
            {
                dfs(i, v);
                low[v] = min(low[v], low[i]);

                if (low[i] >= num[v])
                {
                    ++nBicon;
                    nadj[v].emplace_back(n + nBicon);

                    int vertex;
                    do
                    {
                        vertex = s.back();
                        s.pop_back();

                        nadj[n + nBicon].emplace_back(vertex);
                    } while (vertex != i);
                }
            }
            else

```

```

        low[v] = min(low[v], num[i]);
    }
}

7 String

7.1 Palindrome Tree

// base on idea odd palindrome, even palindrome
// 0-odd is the root of tree
struct node
{
    int len;
    node *child[26], *sufflink;
    node()
    {
        len = 0;
        for (int i = 0; i < 26; ++i)
            child[i] = NULL;
        sufflink = NULL;
    }
};
struct PalindromeTree
{
    node odd, even;
    PalindromeTree()
    {
        odd.len = -1;
        odd.sufflink = &odd;
        even.len = 0;
        even.sufflink = &odd;
    }

    void Assign(string &s)
    {
        node *last = &even;
        for (int i = 0; i < (int)s.size(); ++i)
        {
            node *tmp = last;
            while (s[i - tmp->len - 1] != s[i])
                tmp = tmp->sufflink;
            if (tmp->child[s[i] - 'a'])
            {
                last = tmp->child[s[i] - 'a'];
                continue;
            }
            tmp->child[s[i] - 'a'] = new node;
            last = tmp->child[s[i] - 'a'];
            last->len = tmp->len + 2;
            if (last->len == 1)
            {
                last->sufflink = &even;
                continue;
            }
            tmp = tmp->sufflink;
            while (s[i - tmp->len - 1] != s[i])
                tmp = tmp->sufflink;
            last->sufflink = tmp->child[s[i] - 'a'];
        }
    }
};

```

7.2 Suffix Array

```

// string and array pos start from 0
// but array sa and lcp start from 1

constexpr int N = 3e5 + 5; // change size to size of string;

struct SuffixArray
{
    string s;
    int n, c[N], p[N], rp[N], lcp[N];

    //p[] : suffix array
    // lcp[]: lcp array
}

```



```

void Assign(const string &s)
{
    s = x;
    s.push_back('$'); // Change character here due to range of
                       // charater in string
    n = s.size();
    Build();
    s.pop_back();
    n = s.size();
}

void Build()
{
    vector<int> pn(N), cn(N), cnt(N);

    for (int i = 0; i < n; ++i)
        ++cnt[s[i]];
    for (int i = 1; i <= 256; ++i)
        cnt[i] += cnt[i - 1];
    for (int i = 0; i < n; ++i)
        p[--cnt[s[i]]] = i;

    for (int i = 1; i < n; ++i)
        c[p[i]] = c[p[i - 1]] + (s[p[i]] != s[p[i - 1]]);

    int maxn = c[p[n - 1]];

    for (int i = 0; (1 << i) < n; ++i)
    {
        for (int j = 0; j < n; ++j)
            p[j] = ((p[j] - (1 << i)) % n + n) % n;
        for (int j = 0; j <= maxn; ++j)
            cnt[j] = 0;

        for (int j = 0; j < n; ++j)
            ++cnt[c[p[j]]];

        for (int j = 1; j <= maxn; ++j)
            cnt[j] += cnt[j - 1];

        for (int j = n - 1; ~j; --j)
            pn[--cnt[c[p[j]]]] = p[j];

        for (int j = 1; j < n; ++j)
            cn[pn[j]] = cn[pn[j - 1]] + (c[pn[j]] != c[pn[j - 1]] || c[(pn[j] + (1 << i)) % n] != c[(pn[j - 1] + (1 << i)) % n]);

        maxn = cn[pn[n - 1]];

        for (int j = 0; j < n; ++j)
        {
            p[j] = pn[j];
            c[j] = cn[j];
        }
    }

    void BuildLCP()
    {
        for (int i = 1; i <= n; ++i)
            rp[p[i]] = i;
        for (int i = 0; i < n; ++i)
        {
            if (i)
                lcp[i] = max(lcp[i - 1] - 1, 0);
            if (rp[i] == n)
                continue;

            while (lcp[i] < n - i && lcp[i] < n - p[rp[i] + 1] &&
                    s[i + lcp[i]] == s[p[rp[i] + 1] + lcp[i]])
                ++lcp[i];
        }
    }
}

```

7.3 Suffix Array (Template by Tran Khoi Nguyen)

```

struct suffix_array
{
    vector<int> sa_naive(const vector<int> &s)
    {
        int n = (int)s.size();
        vector<int> sa(n);
        iota(sa.begin(), sa.end(), 0);
        sort(sa.begin(), sa.end(), [&](int l, int r)
        {
            if (l == r) return false;
            for (; l < n && r < n; ++l, ++r) if (s[l] != s[r])
                return s[l] < s[r];
            return l == n; });
        return sa;
    }

    vector<int> sa_doubling(const vector<int> &s)
    {
        int n = (int)s.size();
        vector<int> sa(n), rank = s, tmp(n);
        iota(sa.begin(), sa.end(), 0);
        for (auto k = 1; k < n; k <= 1)
        {
            auto cmp = [&](int x, int y)
            {
                if (rank[x] != rank[y])
                    return rank[x] < rank[y];
                int rx = x + k < n ? rank[x + k] : -1;
                int ry = y + k < n ? rank[y + k] : -1;
                return rx < ry;
            };
            sort(sa.begin(), sa.end(), cmp);
            tmp[sa[0]] = 0;
            for (auto i = 1; i < n; ++i)
                tmp[sa[i]] = tmp[sa[i - 1]] + (cmp(sa[i - 1], sa[i]) ? 1 : 0);
            swap(tmp, rank);
        }
        return sa;
    }

    template <int THRESHOLD_NAIVE = 10, int THRESHOLD_DOUBLING = 40>
    vector<int> sa_is(const vector<int> &s, int sigma)
    {
        int n = (int)s.size();
        if (n == 0)
            return {};
        if (n == 1)
            return {0};
        if (n == 2)
        {
            if (s[0] < s[1])
                return {0, 1};
            else
                return {1, 0};
        }
        if (n < THRESHOLD_NAIVE)
            return sa_naive(s);
        if (n < THRESHOLD_DOUBLING)
            return sa_doubling(s);
        vector<int> sa(n);
        vector<bool> ls(n);
        for (auto i = n - 2; i >= 0; --i)
            ls[i] = (s[i] == s[i + 1]) ? ls[i + 1] : (s[i] < s[i + 1]);
        vector<int> sum_l(sigma), sum_s(sigma);
        for (auto i = 0; i < n; ++i)
        {
            if (!ls[i])
                ++sum_s[s[i]];
            else
                ++sum_l[s[i] + 1];
        }
        for (auto i = 0; i < sigma; ++i)
        {
            sum_s[i] += sum_l[i];
            if (i + 1 < sigma)
                sum_l[i + 1] += sum_s[i];
        }
    }
}

```

```

auto induce = [&](const vector<int> &lms)
{
    fill(sa.begin(), sa.end(), -1);
    vector<int> buf(sigma);
    copy(sum_s.begin(), sum_s.end(), buf.begin());
    for (auto d : lms)
    {
        if (d == n)
            continue;
        sa[buf[s[d]]++] = d;
    }
    copy(sum_l.begin(), sum_l.end(), buf.begin());
    sa[buf[s[n - 1]]++] = n - 1;
    for (auto i = 0; i < n; ++i)
    {
        int v = sa[i];
        if (v >= 1 && !ls[v - 1])
            sa[buf[s[v - 1]]++] = v - 1;
    }
    copy(sum_l.begin(), sum_l.end(), buf.begin());
    for (auto i = n - 1; i >= 0; --i)
    {
        int v = sa[i];
        if (v >= 1 && ls[v - 1])
            sa[--buf[s[v - 1] + 1]] = v - 1;
    }
};

vector<int> lms_map(n + 1, -1);
int m = 0;
for (auto i = 1; i < n; ++i)
    if (!ls[i - 1] && ls[i])
        lms_map[i] = m++;
vector<int> lms;
lms.reserve(m);
for (auto i = 1; i < n; ++i)
    if (!ls[i - 1] && ls[i])
        lms.push_back(i);
induce(lms);
if (m)
{
    vector<int> sorted_lms;
    sorted_lms.reserve(m);
    for (auto v : sa)
        if (lms_map[v] != -1)
            sorted_lms.push_back(v);
    vector<int> rec_s(m);
    int rec_sigma = 0;
    rec_s[lms_map[sorted_lms[0]]] = 0;
    for (auto i = 1; i < m; ++i)
    {
        int l = sorted_lms[i - 1], r = sorted_lms[i];
        int end_l = (lms_map[l] + 1 < m) ? lms[lms_map[l] + 1] : n;
        int end_r = (lms_map[r] + 1 < m) ? lms[lms_map[r] + 1] : n;
        bool same = true;
        if (end_l - 1 != end_r - r)
            same = false;
        else
        {
            for (; l < end_l; ++l, ++r)
                if (s[l] != s[r])
                    break;
            if (l == n || s[l] != s[r])
                same = false;
        }
        if (!same)
            ++rec_sigma;
        rec_s[lms_map[sorted_lms[i]]] = rec_sigma;
    }
    auto rec_sa = sa_is<THRESHOLD_NAIVE, THRESHOLD_DOUBLING>(rec_s, rec_sigma + 1);
    for (auto i = 0; i < m; ++i)
        sorted_lms[i] = lms[rec_sa[i]];
    induce(sorted_lms);
}

return sa;
}

```

```

int n;
// data: sorted sequence of suffices including the empty
// suffix
// rank[i]: position of the suffix i in the suffix array
// lcp[i]: longest common prefix of data[i] and data[i + 1]
// index start from 1
vector<int> data, rank, lcp;
// O(n + sigma)
suffix_array(const vector<int> &s, int sigma) : n((int)s.size()
()), rank(n + 1), lcp(n)
{
    assert(0 <= sigma);
    for (auto d : s)
        assert(0 <= d && d < sigma);
    data = sa_is(s, sigma);
    data.insert(data.begin(), n);
    for (auto i = 0; i <= n; ++i)
        rank[data[i]] = i;
    for (auto i = 0, h = 0; i <= n; ++i)
    {
        if (h > 0)
            --h;
        if (rank[i] == 0)
            continue;
        int j = data[rank[i] - 1];
        for (; j + h <= n && i + h <= n; ++h)
            if ((j + h == n) != (i + h == n) || j + h < n && s
                [j + h] != s[i + h])
                break;
        lcp[rank[i] - 1] = h;
    }
    // O(n log n) time, O(n) space
    template <class T>
    suffix_array(const vector<T> &s, bool prepare_lcp) : n((int)s.
        size()), rank(n + 1), lcp(n)
    {
        vector<int> idx(n);
        iota(idx.begin(), idx.end(), 0);
        sort(idx.begin(), idx.end(), [&](int l, int r)
            { return s[l] < s[r]; });
        vector<int> s2(n);
        int now = 0;
        for (auto i = 0; i < n; ++i)
        {
            if (i && s[idx[i - 1]] != s[idx[i]])
                ++now;
            s2[idx[i]] = now;
        }
        data = sa_is(s2, now + 1);
        data.insert(data.begin(), n);
        for (auto i = 0; i <= n; ++i)
            rank[data[i]] = i;
        for (auto i = 0, h = 0; i <= n; ++i)
        {
            if (h > 0)
                --h;
            if (rank[i] == 0)
                continue;
            int j = data[rank[i] - 1];
            for (; j + h <= n && i + h <= n; ++h)
                if ((j + h == n) != (i + h == n) || j + h < n && s
                    [j + h] != s[i + h])
                    break;
            lcp[rank[i] - 1] = h;
        }
    }
    // RMQ must be built over lcp
    // O(1)
    template <class RMQ>
    int longest_common_prefix(int i, int j, const RMQ &rmq) const
    {
        assert(0 <= i && i <= n && 0 <= j && j <= n);
        return i == j ? n - i : rmq.query(min(rank[i], rank[j]),
            max(rank[i], rank[j]));
    }
};

```

7.4 Aho Corasick - Extended KMP

```

constexpr int ALPHABET_SIZE = 26;
constexpr int firstCharacter = 'a';

struct Node
{
    Node *to[ALPHABET_SIZE];
    Node *suflink;
    int ending_length; // 0 if is not ending

    Node()
    {
        for (int i = 0; i < ALPHABET_SIZE; ++i)
            to[i] = NULL;
        suflink = NULL;
        ending_length = false;
    }
};

struct AhoCorasick
{
    Node *root;

    AhoCorasick()
    {
        root = new Node();
    }

    void add(const string &s)
    {
        Node *cur_node = root;

        for (char c : s)
        {
            int v = c - firstCharacter;

            if (!cur_node->to[v])
                cur_node->to[v] = new Node();

            cur_node = cur_node->to[v];

            cur_node->ending_length = s.size();
        }

        // if a->to[v] == NULL
        // for convenient a->to[v] = the node x->to[v] that a match x
        // and x->to[v] != NULL
        // root -> suflink = root

        void build()
        {
            queue<Node *> Q;
            root->suflink = root;
            Q.push(root);

            while (!Q.empty())
            {
                Node *par = Q.front();
                Q.pop();
                for (int c = 0; c < ALPHABET_SIZE; ++c)
                {
                    if (par->to[c])
                    {
                        par->to[c]->suflink = par == root ? root : par
                            ->suflink->to[c];
                        Q.push(par->to[c]);
                    }
                    else
                    {
                        par->to[c] = par == root ? root : par->suflink
                            ->to[c];
                    }
                }
            }
        }
    }
};

```

};

7.5 Aho Corasick (Template by Tran Khoi Nguyen)

```

struct aho_corasick
{
    struct tk
    {
        ll link;
        ll nxt[27];
        ll par;
        char ch;
        ll go[27];
        ll val;
        ll leaf;
        tk(ll par = -1, char ch = 'a') : par(par), ch(ch)
        {
            memset(nxt, -1, sizeof(nxt));
            memset(go, -1, sizeof(go));
            val = -1;
            link = -1;
            leaf = 0;
        }
    };

    vector<tk> vt;
    void init()
    {
        vt.clear();
        vt.pb({-1, 'a'});
    }

    ll add(string s, ll val)
    {
        ll nw = 0;
        for (auto to : s)
        {
            if (vt[nw].nxt[to - 'a' + 1] == -1)
            {
                vt[nw].nxt[to - 'a' + 1] = vt.size();
                vt.pb({nw, to});
            }
            nw = vt[nw].nxt[to - 'a' + 1];
        }
        vt[nw].leaf = val;
        return nw;
    }

    ll get_val(ll u)
    {
        if (u == 0)
            return 0;
        if (vt[u].val == -1)
        {
            vt[u].val = vt[u].leaf + get_val(get_link(u));
        }
        return vt[u].val;
    }

    ll go(ll v, ll t)
    {
        if (vt[v].go[t] == -1)
        {
            if (vt[v].nxt[t] != -1)
            {
                vt[v].go[t] = vt[v].nxt[t];
            }
            else
            {
                if (v == 0)
                    vt[v].go[t] = 0;
                else
                    vt[v].go[t] = go(get_link(v), t);
            }
        }
        return vt[v].go[t];
    }

    ll get_link(ll v)
    {
        if (vt[v].link == -1)
        {

```

```

    if (vt[v].par == 0 || v == 0)
        vt[v].link = 0;
    else
        vt[v].link = go(get_link(vt[v].par), vt[v].ch - 'a' + 1);
}
return vt[v].link;
}
ll get(string s)
{
    ll nw = 0;
    ll ans = 0;
    for (auto to : s)
    {
        nw = go(nw, to - 'a' + 1);
        ans += get_val(nw);
    }
    return ans;
}
};

```

7.6 Suffix Tree (Template by Tran Khoi Nguyen)

```

struct tk
{
    map<ll, ll> nxt;
    ll par, f, len;
    ll link;
    tk(ll par = -1, ll f = 0, ll len = 0) : par(par), f(f), len(len)
    {
        nxt.clear();
        link = -1;
    }
};

struct Suffix_Tree
{
    vector<tk> st;
    ll node;
    ll dis;
    s
    ll n;
    vector<ll> s;
    void init()
    {
        st.clear();
        node = 0;
        dis = 0;
        st.emplace_back(-1, 0, base);
        n = 0;
    }

    void go_edge()
    {
        while (dis > st[st[node].nxt[s[n - dis]]].len)
        {
            node = st[node].nxt[s[n - dis]];
            dis -= st[node].len;
        }
    }

    void add_char(ll c)
    {
        ll last = 0;
        s.pb(c);
        n = s.size();
        dis++;
        while (dis > 0)
        {
            go_edge();
            ll edge = s[n - dis];
            ll &v = st[node].nxt[edge];
            ll t = s[st[v].f + dis - 1];
            if (v == 0)
            {
                v = st.size();

```

```

                st.emplace_back(node, n - dis, base);
                st[last].link = node;
                last = 0;
            }
            else if (c == t)
            {
                st[last].link = node;
                return;
            }
            else
            {
                ll u = st.size();
                st.emplace_back(node, st[v].f, dis - 1);
                st[u].nxt[c] = st.size();
                st.emplace_back(u, n - 1, base);
                st[u].nxt[t] = v;
                st[v].f += (dis - 1);
                st[v].len -= (dis - 1);
                v = u;
                st[last].link = u;
                last = u;
            }
        }
        if (node == 0)
            dis--;
        else
            node = st[node].link;
    }
};

```

7.7 Z Function

```

// string start from 1
// f[i] = longest prefix match with s[i...i + f[i] - 1]

constexpr int N = 2e5 + 5;

void Build(string &s, int n, int f[N]) // n = size of string, f = z array
{
    int l(1), r(1);

    f[1] = n;

    for (int i = 2; i <= n; ++i)
        if (r < i)
        {
            l = r = i;
            while (r <= n && s[r - i + 1] == s[r])
                ++r;
            f[i] = r - i;
            --r;
        }
        else if (f[i - 1 + 1] < r - i + 1)
            f[i] = f[i - 1 + 1];
        else
        {
            l = i;
            while (r <= n && s[r - i + 1] == s[r])
                ++r;
            f[i] = r - i;
            --r;
        }
}

```

8 Data structures

8.1 Ordered Set

```

#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;

#define ordered_set tree<int, null_type, less<int>, rb_tree_tag,
tree_order_statistics_node_update>

```

```

/*
order_of_key(k) : Number of items strictly smaller than k.
find_by_order(k) : K-th element in a set (counting from zero).
*/

8.2 Fenwick Tree (With Walk on tree)

// This is equivalent to calculating lower_bound on prefix sums
array
// LOGN = log2(N)

struct FenwickTree
{
    int n, LOGN;
    ll a[N]; // BIT array

    FenwickTree()
    {
        memset(a, 0, sizeof a);
    }

    void Update(int p, ll v)
    {
        for (; p <= n; p += p & -p)
            a[p] += v;
    }

    ll Get(int p)
    {
        ll ans(0);

        for (; p; p -= p & -p)
            ans += a[p];

        return ans;
    }

    int search(ll v)
    {
        ll sum = 0;
        int pos = 0;
        for (int i = LOGN; i >= 0; i--)
        {
            if (pos + (1 << i) <= n && sum + a[pos + (1 << i)] < v)
            {
                sum += a[pos + (1 << i)];
                pos += (1 << i);
            }
        }
        return pos + 1;
        //+1 because pos will be position of largest value less than v
    }
};

8.3 Convex Hull Trick (Min)

// If you want to get maximum, sort coef (A) not decreasing and
change B[line.back()] > B[i] into B[line.back()] < B[i]

struct ConvexHullTrick
{
    vector<ll> A, B;
    vector<int> line;
    vector<ld> point;
    ConvexHullTrick(int n = 0)
    {
        A.resize(n + 2, 0);
        B.resize(n + 2, 0);
        point.emplace_back(-Inf);
    }

    ld ff(int x, int y)
    {
        return (ld)1.0 * (B[y] - B[x]) / (A[x] - A[y]);
    }
};

```

```

}

void Add(int i)
{
    while ((int)line.size() > 1 || ((int)line.size() == 1 && A[line.back()] == A[i]))
    {
        if (A[line.back()] == A[i])
        {
            if (B[line.back()] > B[i])
            {
                line.pop_back();
                if (!line.empty())
                    point.pop_back();
            }
            else
                break;
        }
        else
        {
            if (ff(i, line.back()) <= ff(i, line[line.size() - 2]))
            {
                line.pop_back();
                if (!line.empty())
                    point.pop_back();
            }
            else
                break;
        }
    }

    if (line.empty() || A[line.back()] != A[i])
    {
        if (!line.empty())
            point.emplace_back(ff(line.back(), i));
        line.emplace_back(i);
    }
}

ll Get(int x)
{
    int j = lower_bound(point.begin(), point.end(), x) - point.begin();
    return A[line[j - 1]] * x + B[line[j - 1]];
}
};

```

8.4 Dynamic Convex Hull Trick (Min)

```

struct Line
{
    mutable ll k, m, p;
    bool operator<(const Line& o) const
    {
        if (k==o.k) return m>o.m;
        return k > o.k;
    }
    bool operator<(ll x) const
    {
        return p < x;
    }
};

struct LineContainer : multiset<Line, less<>>
{
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b)
    {
        return a / b - ((a ^ b) < 0 && a % b);
    }
    bool isect(iterator x, iterator y)
    {
        if (y == end())
            return x->p = inf, 0;
        if (x->k == y->k)
            x->p = x->m < y->m ? inf : -inf;
        else
            x->p = div(y->m - x->m, x->k - y->k);
    }
};

```

```

        return x->p >= y->p;
    }
    void add(ll k, ll m)
    {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z))
            z = erase(z);
        if (x != begin() && isect(--x, y))
            isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p)
            isect(x, erase(y));
    }
    ll query(ll x)
    {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};

```

8.5 SPlay Tree

```

struct KNode
{
    int Value;
    int Size;
    KNode *P, *L, *R;
};
using QNode = KNode *;
KNode No_thing_here;
QNode nil = &No_thing_here, root;

void Link(QNode par, QNode child, bool Right)
{
    child->P = par;
    if (Right)
        par->R = child;
    else
        par->L = child;
}

void Update(QNode &a)
{
    a->Size = a->L->Size + a->R->Size + 1;
}

void Init()
{
    nil->Size = 0;
    nil->P = nil->L = nil->R = nil;
    root = nil;
    for (int i = 1; i <= n; ++i)
    {
        QNode cur = new KNode;
        cur->P = cur->L = cur->R = nil;
        cur->Value = i;
        Link(cur, root, false);
        root = cur;
        Update(root);
    }
}

void Rotate(QNode x)
{
    QNode y = x->P;
    QNode z = y->P;
    if (x == y->L)
    {
        Link(y, x->R, false);
        Link(x, y, true);
    }
    else
    {
        Link(y, x->L, true);
        Link(x, y, false);
    }
    Update(y);
}

```

```

Update(x);
x->P = nil;
if (z != nil)
    Link(z, x, z->R == y);
}

void Up_to_Root(QNode x)
{
    while (1) {
        QNode y = x->P;
        QNode z = y->P;
        if (y == nil)
            break;
        if (z != nil) {
            if ((x == y->L) == (y == z->L))
                Rotate(y);
            else
                Rotate(x);
        }
        Rotate(x);
    }
}

QNode The_kth(QNode x, int k)
{
    while (true)
    {
        if (x->L->Size == k - 1)
            return x;
        if (x->L->Size >= k)
            x = x->L;
        else
        {
            k -= x->L->Size + 1;
            x = x->R;
        }
    }
    return nil;
}

void Split(QNode x, int k, QNode &a, QNode &b)
{
    if (k == 0)
    {
        a = nil;
        b = x;
        return;
    }
    QNode cur = The_kth(x, k);
    Up_to_Root(cur);
    a = cur;
    b = a->R;
    a->R = nil;
    b->P = nil;
    Update(a);
}

QNode Join(QNode a, QNode b)
{
    if (a == nil)
        return b;
    while (a->R != nil)
        a = a->R;
    Up_to_Root(a);
    Link(a, b, true);
    Update(a);
    return a;
}

void Print(QNode &a)
{
    if (a->L != nil)
        Print(a->L);
    cout << (a->Value) << " ";
    if (a->R != nil)
        Print(a->R);
}

```

8.6 Hashing (Template by Tran Khoi Nguyen)

```

struct Hashing
{
    vector<vector<vector<ll>>> f;
    vector<ll> mod;
    vector<vector<ll>>> mu;
    vector<ll> chr;
    ll num;
    ll base;
    void init()
    {
        num = 2;
        f.clear();
        mod.clear();
        mu.clear();
        chr.clear();
        vector<ll> vt = {999244369, 999254351, 999154309,
            989154311, 989254411, 997254397, 991294387,
            991814399, 994114351, 994914359, 994024333};
        random_shuffle(vt.begin(), vt.end());
        base = 317;
        for (int i = 1; i <= 26; i++)
            chr.pb(abs((ll)(rnd())));
        for (int i = 0; i < num; i++)
        {
            f.emplace_back();
            mod.pb(vt[i]);
            vector<ll> pt;
            pt.pb(1);
            for (int j = 1; j < maxn; j++)
            {
                pt.pb((pt.back() * base) % mod[i]);
            }
            mu.pb(pt);
        }
    }
    ll add(string s)
    {
        ll n = s.length();
        ll id = f[0].size();
        for (int j = 0; j < num; j++)
        {
            vector<ll> vt1;
            vt1.pb(0);
            for (int i = 1; i <= n; i++)
            {
                vt1.pb((vt1.back() * base + chr[s[i - 1] - 'a']) % mod[j]);
            }
            f[j].pb(vt1);
        }
        return id;
    }
    pll get_hash(ll id, ll l, ll r)
    {
        return make_pair(((f[0][id][r] - f[0][id][l - 1] * mu[0][r - l + 1]) % mod[0] + mod[0]) % mod[0], ((f[1][id][r] - f[1][id][l - 1] * mu[1][r - l + 1]) % mod[1] + mod[1]) % mod[1]);
    }
};

```

8.7 BIT 2D (Template by Tran Khoi Nguyen)

```

struct BIT_2D
{
    vector<ll> vt;
    vector<vector<ll>> f;
    vector<vector<ll>> node;
    void init(vector<ll> vt1)
    {
        vt = vt1;
        sort(vt.begin(), vt.end());
        vt.resize(unique(vt.begin(), vt.end()) - vt.begin());
        f = vector<vector<ll>>(vt.size() + 2, vector<ll>(0));
        node = vector<vector<ll>>(vt.size() + 2, vector<ll>(0));
    }
    void rearrange()
    {
        for (int i = 1; i <= vt.size(); i++)
        {
            sort(node[i].begin(), node[i].end());
            node[i].resize(unique(node[i].begin(), node[i].end()) - node[i].begin());
            f[i] = vector<ll>(node[i].size() + 2, 0);
        }
    }
    void fake_update(ll x, ll y)
    {
        for (int i = lower_bound(vt.begin(), vt.end(), x) - vt.begin() + 1; i <= vt.size(); i += i & (-i))
        {
            node[i].pb(y);
        }
    }
    void fake_get(ll x, ll y)
    {
        for (int i = lower_bound(vt.begin(), vt.end(), x) - vt.begin() + 1; i <= vt.size(); i -= i & (-i))
        {
            node[i].pb(y);
        }
    }
    void update(ll x, ll y, ll val)
    {
        for (int i = lower_bound(vt.begin(), vt.end(), x) - vt.begin() + 1; i <= vt.size(); i += i & (-i))
        {
            for (int j = lower_bound(node[i].begin(), node[i].end(), y) - node[i].begin() + 1; j <= node[i].size(); j += j & (-j))
            {
                f[i][j] = f[i][j] + val;
            }
        }
    }
    ll get(ll x, ll y)
    {
        ll ans = 0;
        for (int i = lower_bound(vt.begin(), vt.end(), x) - vt.begin() + 1; i <= vt.size(); i -= i & (-i))
        {
            for (int j = lower_bound(node[i].begin(), node[i].end(), y) - node[i].begin() + 1; j <= node[i].size(); j -= j & (-j))
            {
                ans += f[i][j];
            }
        }
        return ans;
    }
};

```

8.8 Wavelet Tree

```

struct wavelet_tree{
    int lo, hi;
    wavelet_tree *l, *r;
    vector<int> b;

    //nos are in range [x,y]
    //array indices are [from, to]
    wavelet_tree(int *from, int *to, int x, int y){
        lo = x, hi = y;
        if(lo == hi or from >= to) return;
        int mid = (lo+hi)/2;
        auto f = [mid](int x){
            return x <= mid;
        };
        b.reserve(to-from+1);
        b.push_back(0);
        for(auto it = from; it != to; it++)
            b.push_back(b.back() + f(*it));
        //see how lambda function is used here
        auto pivot = stable_partition(from, to, f);
        l = new wavelet_tree(from, pivot, lo, mid);
        r = new wavelet_tree(pivot, to, mid+1, hi);
    }

    //kth smallest element in [l, r]
    int kth(int l, int r, int k){
        if(l > r) return 0;
        if(lo == hi) return lo;
        int inLeft = b[r] - b[l-1];
        int lb = b[l-1]; //amt of nos in first (l-1) nos
                        //that go in left
        int rb = b[r]; //amt of nos in first (r) nos that
                        //go in left
        if(k <= inLeft) return this->l->kth(lb+1, rb, k);
        return this->r->kth(l-lb, r-rb, k-inLeft);
    }

    //count of nos in [l, r] Less than or equal to k
    int LTE(int l, int r, int k) {
        if(l > r or k < lo) return 0;
        if(hi <= k) return r - l + 1;
        int lb = b[l-1], rb = b[r];
        return this->l->LTE(lb+1, rb, k) + this->r->LTE(l-lb, r-rb, k);
    }

    //count of nos in [l, r] equal to k
    int count(int l, int r, int k) {
        if(l > r or k < lo or k > hi) return 0;
        if(lo == hi) return r - l + 1;
        int lb = b[l-1], rb = b[r], mid = (lo+hi)/2;
        if(k <= mid) return this->l->count(lb+1, rb, k);
        return this->r->count(l-lb, r-rb, k);
    }
};

/**
 * Example use:
 * int a[N];
 * wavelet_tree T(a+1, a+n+1, 1, MAX);
 */

```